

**EPANAW BAGOBO: AN AI-INTEGRATED WEBSITE FRAMEWORK FOR
THE PRESERVATION AND REVITALIZATION OF THE BAGOBO
TAGABAWA DIALECT AND CULTURAL HERITAGE**

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**CAPSTONE PROJECT OUTLINE SUBMITTED TO THE FACULTY OF
THE COLLEGE OF INFORMATION AND DIGITAL SCIENCES
(CIDS), DAVAO DEL SUR STATE COLLEGE,
MATTI, DIGOS CITY, DAVAO DEL SUR
IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS
FOR THE DEGREE OF**

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

MARCH 2026

CHAPTER I

INTRODUCTION

Background of the Study

Cultural preservation in the Philippines is strongly supported by national policies and institutions tasked with safeguarding the country's diverse heritage. The National Cultural Heritage Act (RA 10066) provides a legal framework for the protection and conservation of cultural properties, while agencies such as the National Commission for Culture and the Arts and the National Museum of the Philippines implement programs that promote indigenous traditions, languages, and historical sites. These efforts include documentation, conservation of artifacts, and educational campaigns aimed at raising public awareness. However, despite these initiatives, challenges such as rapid urbanization, globalization, and limited resources continue to threaten cultural heritage. In recent years, the integration of digital technologies has emerged as a promising approach to preservation. Digital platforms, online repositories, and information systems now play a crucial role in organizing, storing, and disseminating cultural knowledge, enabling wider access and long-term sustainability. Scholarly studies on digital preservation emphasize that technology, when combined with community participation, enhances the protection and transmission of indigenous knowledge systems.

The Philippines is home to more than 175 ethnolinguistic groups, reflecting its rich cultural diversity and deep historical roots. Among these groups are the Bagobo people

of southern Mindanao, composed of subgroups such as the Tagabawa, Clata (Guiangan), and Ubo. These communities share a common identity yet maintain distinct dialects, traditions, and artistic expressions. Historically, the Bagobo have inhabited areas surrounding the Davao Gulf and the slopes of Mount Apo, where their cultural practices have been preserved through oral traditions and community life. However, globalization and modernization have significantly influenced cultural dynamics, leading to shifts in language use and lifestyle, particularly among the younger generation. This study, which proposes the development of a comprehensive website framework for the preservation and revitalization of the Bagobo dialect and cultural heritage, responds to the growing need for innovative solutions. Existing research supports the use of digital platforms as effective tools for cultural education, engagement, and preservation, highlighting their ability to connect traditional knowledge with modern audiences while promoting inclusivity and accessibility.

Despite the presence of legal frameworks and cultural programs, the Bagobo language and cultural heritage are increasingly at risk of decline. Factors such as migration, exposure to dominant cultures, and the preference for widely used languages have contributed to the gradual loss of indigenous linguistic practices. Younger generations, in particular, are becoming less familiar with their native dialect, leading to a weakening of cultural identity and continuity. Furthermore, there is a lack of centralized, accessible, and interactive platforms dedicated to preserving and teaching the Bagobo dialect and traditions. Existing resources are often scattered, limited in scope, or not easily accessible to the community. This gap highlights a critical problem: without

effective and sustainable preservation strategies, the Bagobo language and its associated cultural knowledge may continue to diminish, potentially leading to cultural erosion over time.

In response to these challenges, this study proposes the development of a digital website platform that serves as both a repository and an interactive learning environment for the Bagobo dialect and cultural heritage. The proposed system aims to integrate features such as multimedia content, language tutorials, and community participation tools to encourage active engagement and knowledge sharing. This initiative aligns with the objectives of the Indigenous Peoples' Rights Act (IPRA) 8371, which emphasizes the protection and promotion of indigenous cultural rights and identity. By leveraging technology, the platform seeks to ensure the sustainable preservation of cultural resources while fostering intergenerational transmission of knowledge. Ultimately, the study envisions creating an inclusive, accessible, and community-centered solution that not only preserves the Bagobo heritage but also strengthens cultural pride and identity in the modern digital era.

Objectives of the Study

This study aims to design and develop a web-based system for the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage through digital archiving, interactive learning, and community-driven technologies.

Specifically, it aims to:

1. Design and develop a web-based cultural preservation system.

1.1 Implement a role-based access control mechanism that defines permissions for Super Admin, Admin, and User accounts.

1.2 Super admin controls the whole system

1.3 Admin manages the content; and

1.4 User access the learning materials.

2. Develop a centralized digital repository for the structured storage and retrieval of Bagobo cultural resources

2.1 audio,

2.2 video,

2.3 textual, and;

2.4 image data.

3. Incorporate an interactive learning management module.

3.1 Delivers lessons, assessments, and user engagement activities;

3.2 Integrate an AI-driven component like chatbot to provide adaptive learning support and user assistance.

4. Determine the system's effectiveness in facilitating the preservation, accessibility, and dissemination of the Bagobo dialect and cultural heritage through digital means.

5. Evaluate the developed system based on the following software quality attributes:

3.1 Functionality;

3.2 Usability; and

3.3 Reliability.

Scope and Limitation

This study focuses on the development of a web-based cultural preservation system for the Bagobo Tagabawa community, aimed at documenting and promoting their language, traditions, and cultural heritage through a centralized digital platform. The system includes features such as a digital repository, interactive learning modules, role-based access control, and an AI-assisted learning component. It is designed to be user-friendly and accessible, allowing both community members and learners to engage with cultural content. While primarily focused on the Bagobo Tagabawa culture, the system also has potential for future expansion to other indigenous communities.

However, the study is limited by several factors, including restricted internet access and device availability in rural areas, which may affect user participation. The AI functionality is limited to basic assistance features and does not include advanced artificial intelligence capabilities. Additionally, the system's content is dependent on available data collected during development, and continuous updates require further support.

Evaluation is also limited to selected software quality attributes and respondents, while long-term maintenance and scalability are beyond the scope of the study.

Significance of the Study

This study is significant as it contributes to the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage through a web-based digital platform. It provides a centralized system for documenting, storing, and accessing cultural resources, helping prevent the loss of indigenous knowledge caused by modernization and globalization. It is very significant to this following:

Bagobo Community. The system serves as a digital archive and learning tool that helps preserve their language, traditions, oral narratives, and cultural identity. It also promotes intergenerational knowledge transfer by allowing elders to share cultural knowledge and enabling younger generations to learn and engage with their heritage in an interactive and meaningful way.

Students. The system provides a centralized and organized source of cultural and linguistic data that can be used for academic studies, cultural research, and language learning. It simplifies access to information that is often scattered or difficult to obtain, thereby supporting educational and research activities related to indigenous culture.

Educational Institutions. The study offers a supplementary learning resource that can be integrated into cultural education programs, promoting awareness and

appreciation of indigenous heritage among learners. It also supports innovative teaching approaches through interactive and technology-based learning tools.

LGU's and Cultural Organizations. The system provides a potential tool for cultural preservation initiatives and community development programs. It supports the objectives of safeguarding indigenous identity and can assist in policy implementation related to cultural conservation.

Future Researchers and Developers. This study serves as a reference for creating similar systems that combine cultural preservation with web-based technologies and AI integration. It provides a foundation for further improvements and expansion to other indigenous communities and cultural contexts.

Definition of Terms

This study is composed of terminologies that are commonly used particularly in the context of this study.

Artificial Intelligence (AI) – A system feature that uses intelligent algorithms (e.g., chatbot or recommendation system) to provide adaptive learning support and user assistance.

Admin – A user role responsible for managing and uploading content, monitoring system activity, and maintaining the platform.

Adaptive Learning – A learning approach where the system adjusts content and difficulty based on the user's performance and progress.

Bagobo - One of the indigenous ethnolinguistic groups in Mindanao, Philippines, known for their rich cultural traditions, oral literature, rituals, and distinct dialect.

Bagobo Tagabawa – one of the distinct dialects that the Bagobo indigenous group has.

Chatbot Tutor – An AI-based tool designed to interact with users, answer questions, provide explanations, and assist in learning the Bagobo dialect.

Cultural Heritage – The traditions, customs, language, beliefs, and material expressions inherited from previous generations that define the identity of a community.

Digital Preservation – The use of digital technologies to collect, store, manage, and safeguard cultural and historical resources for long-term accessibility and protection.

Digital Repository – A structured online storage system used for organizing and managing digital files such as documents, images, audio recordings, and videos for easy access and retrieval.

Web-Based System – A software application that runs on a web browser and can be accessed through the internet or local network without requiring installation on a device.

Role-Based Access Control – A security framework that restricts system access based on user roles and permissions to ensure proper data management and system security.

Super Admin – The highest-level user with full administrative control over the system, including user management, system configuration, and content approval.

User – An end-user who accesses the system primarily for learning, viewing cultural content, and interacting with the learning modules.

Interactive Learning Module – A feature of the system that provides structured educational content such as lessons, quizzes, and activities to enhance user engagement and learning experience.

Repository Management – The process of organizing, uploading, updating, and maintaining digital content within the system's database.

User Interface (UI) – The visual layout and design of the system that allows users to interact with its features and functions.

User Experience (UX) – The overall experience of users when interacting with the system, including ease of use, accessibility, and satisfaction.

System – A structured set of interconnected components (hardware, software, and processes) that work together to perform specific functions, such as managing data, processing information, and delivering services to users within a defined environment.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Related Literature

This chapter reviews recent studies related to digital preservation, web-based systems, artificial intelligence, and data management as foundations for developing the proposed cultural preservation system. It highlights how these technologies support the documentation, accessibility, and protection of cultural heritage, guiding the design and implementation of the study.

Literature Map of the Study

The literature map of this study illustrates the thematic evolution and interconnectedness of research that supports the design, development, and security of the Web-Based Bagobo Tagabawa System. By synthesizing diverse academic perspectives, the map establishes a theoretical framework that justifies the integration of modern digital infrastructure as seen in Diagram 1.

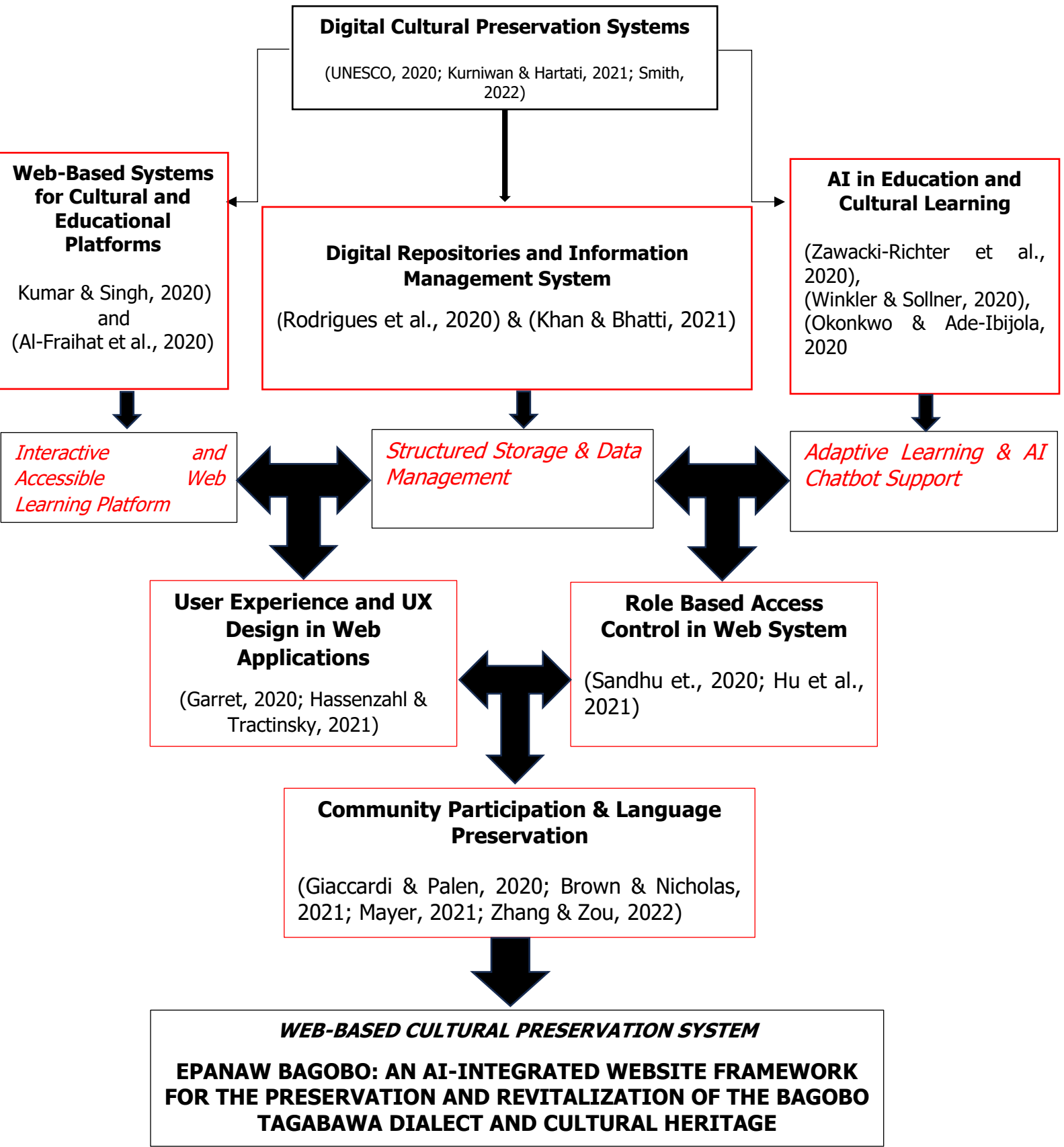


Diagram 1: Literature Map of the Study

Digital Cultural Preservation System

Digital preservation has become an essential strategy in safeguarding indigenous knowledge and cultural heritage in the modern era. With the increasing impact of globalization, many indigenous languages and traditions are at risk of being lost, making digital documentation a necessary intervention.

According to UNESCO (2020), digital technologies play a significant role in preserving intangible cultural heritage by enabling systematic documentation, storage, and wider dissemination of cultural materials.

Similarly, Kurniawan and Hartati (2021) emphasized that digital archiving systems ensure the long-term sustainability of cultural resources such as oral traditions, language recordings, and historical documents.

Furthermore, Smith (2022) noted that digital preservation enhances the visibility and accessibility of indigenous cultures, although it requires active community participation to maintain authenticity and cultural sensitivity. These findings highlight the importance of developing a digital platform that not only preserves cultural heritage but also ensures its accessibility for future generations.

Web-Based Systems for Cultural and Educational Platforms

Web-based systems have become widely utilized in both educational and cultural preservation initiatives due to their accessibility, scalability, and flexibility. These systems

allow users to access information anytime and anywhere, making them highly effective tools for knowledge dissemination.

Kumar and Singh (2020) stated that web-based platforms provide efficient and scalable solutions for delivering content, particularly in educational environments.

In addition, Al-Fraihat et al. (2020) found that the success of web-based learning systems is largely influenced by system quality, information quality, and service quality, which directly impact user satisfaction and engagement. Their study further emphasized that interactive features and well-designed interfaces enhance user experience and learning outcomes. These findings support the use of a web-based system in cultural preservation, as it enables wider access and promotes user engagement in learning indigenous culture.

Digital Repositories and Information Management Systems

Digital repositories play a crucial role in managing and preserving large volumes of data in an organized and accessible manner. These systems provide structured storage, efficient retrieval, and long-term preservation of digital content.

According to Rodrigues et al. (2020), digital repositories utilize features such as metadata tagging, indexing, and search functionalities to improve data accessibility and management.

Similarly, Khan and Bhatti (2021) highlighted that centralized repository systems enhance knowledge sharing and ensure data consistency, which are essential for sustainable information management. Their study also emphasized that repositories are particularly valuable in preserving cultural and academic resources. These findings justify the integration of a centralized digital repository in the proposed system to ensure the efficient storage and retrieval of Bagobo cultural materials.

AI in Education and Cultural Learning

Artificial Intelligence (AI) has significantly transformed educational systems by introducing adaptive and interactive learning experiences. AI-powered tools, such as chatbots and recommendation systems, have been widely used to enhance user engagement and personalize learning.

Zawacki-Richter et al. (2020) explained that AI enables adaptive learning by analyzing user behavior and adjusting content according to individual needs and performance.

Furthermore, Winkler and Söllner (2020) found that chatbot-based learning systems improve interaction by providing instant responses and continuous support to users.

In addition, Okonkwo and Ade-Ibijola (2021) concluded that AI chatbots are particularly effective in language learning, as they simulate conversational interactions and provide real-time feedback. These studies support the integration of an AI-driven

component in the system to enhance the learning experience and promote engagement in cultural education.

User Experience and UX Design in Web Applications

User experience (UX) design is critical in determining the success and usability of web-based systems. According to Garrett (2020), a well-designed user interface enhances usability by making systems intuitive, efficient, and easy to navigate.

In addition, a study by Hassenzahl and Tractinsky (2021) emphasized that positive user experience increases user satisfaction and engagement, which are essential for system adoption. These findings suggest that the proposed system must prioritize user-friendly design to ensure accessibility for both technical and non-technical users.

Role Based Access Control in Web System

Security and proper access management are essential components of web-based systems, especially those handling sensitive or valuable data. Role-Based Access Control (RBAC) is a widely used security model that restricts system access based on user roles and permissions.

According to Sandhu et al. (2020), RBAC ensures that users can only access information relevant to their roles, thereby improving system security and efficiency.

Similarly, Hu et al. (2021) emphasized that RBAC reduces the risk of unauthorized access and simplifies system administration by clearly defining user responsibilities. This model is particularly effective in systems with multiple user roles, such as administrators and general users. These findings support the implementation of RBAC in the proposed system to ensure secure and organized management of user access.

Community Participation & Language Preservation

Community involvement is a key factor in the success of cultural preservation initiatives. According to Giaccardi and Palen (2020), participatory systems that involve local communities in content creation and management ensure authenticity and sustainability. Indigenous knowledge is best preserved when elders and cultural custodians actively contribute to the documentation process.

Similarly, a study by Brown and Nicholas (2021) emphasized that community-driven digital platforms empower indigenous groups by giving them control over how their culture is represented. These findings highlight the importance of incorporating user participation features in the system to encourage collaboration and cultural ownership.

The reviewed literature shows a strong connection between digital technology and cultural preservation. Digital repositories and web-based systems provide efficient platforms for storing and sharing cultural data, while AI technologies enhance user

engagement and learning. This study builds upon these concepts to address the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Related Studies

This section presents related studies and existing systems that focus on Indigenous language preservation, digital heritage documentation, and interactive learning technologies.

List of the Systems Related to the Study

This section presents existing systems related to language preservation and cultural documentation. It serves as a basis for identifying useful features and gaps for the proposed study as shown in Table 1.

Table 1. Comparative Analysis of Indigenous Preservation Systems and Projects

Name of System/Project	Author/Year Cite	Description	Strength	Weaknesses	Application
Mukurtu CMS	Christen et al. (2017)	An open-source content management system designed specifically for indigenous communities to manage digital heritage	Cultural Protocols: Features granular access levels (e.g., gender-specific, sacred knowledge) that respect community traditions.	Complexity: Requires significant technical expertise to set up and maintain; steep learning curve for non-	The strict access control model will be adopted to protect sensitive Bagobo rituals that should not be publicly viewable.

		with cultural protocols.		technical elders.	
Marayum	UP Diliman & DOST-ASTI (2021)	A community-built online dictionary platform for Philippine indigenous languages that allows crowdsourced contributions.	Community-Led: Empowering native speakers to be content creators and editors ensures linguistic accuracy and ownership.	Scope Limitation: Primarily text-based (dictionary focus); lacks broader cultural context like oral history or multimedia storytelling.	The crowdsourcing feature will be utilized to allow Bagobo youth and elders to continuously update the dialect dictionary.
Higaonon Mobile Dictionary	MDPI / Local Researchers (2025)	A mobile application developed to preserve the Higaonon language in Mindanao through digitized vocabulary and audio.	Offline Accessibility: Designed to function without internet, addressing the "digital divide" in rural Mindanao areas.	Platform Dependency: limited to mobile devices; lacks a centralized web archive for larger historical documents.	The "Offline Mode" architecture is critical for the Bagobo framework to ensure usability in remote areas of Bansalan.
FirstVoices	First Peoples' Cultural Council (2013)	A web-based suite of tools for archiving language data, including keyboards, dictionaries,	Rich Media Support: Excellent integration of high-quality audio and video for pronunciation and storytelling.	Cost & Scale: High resource requirement for server maintenance and data curation.	Serves as a model for the "Digital Repository" component, emphasizing the need for high-quality audio recording standards.

		s, and stories for Canadian First Nations.			
Irula Language MALL	Cureus Journal (2025)	A Mobile-Assisted Language Learning (MALL) app using gamification to teach the Irula tribal language to youth.	Youth Engagement: Uses gamification (quizzes, levels) to make language learning attractive to the younger generation.	Narrow Focus: Focuses heavily on vocabulary drills rather than deep cultural immersion or history.	The interactive learning portal will incorporate gamified elements (badges, quizzes) to combat linguistic assimilation among Bagobo youth.
Respectful Design Framework	Reitsma et al. (2013)	A participatory design approach that prioritizes "ownership of process" over just the final technological product.	Ethical Foundation: Ensures the community feels ownership of the system, preventing it from feeling like an external academic imposition.	Abstract Nature: It is a methodology rather than a software tool; requires translation into concrete technical requirements.	This framework will guide the "Community Engagement Interface," ensuring Bagobo elders have veto power over design decisions.
Word Weaver	NRC Canada (2019)	An open-source tool specifically designed to conjugate complex verbs in indigenous languages.	Linguistic Depth: Addresses complex grammatical structures often ignored by simple word-translation dictionaries	Niche Utility: Very specific usage; does not manage general cultural assets like photos or history.	Useful for the advanced modules of the E-Learning portal to teach the complexities of Bagobo sentence structure.

Digital Preservation of IK Framework	Biyela et al. (2016)	A comparative framework for digitizing Indigenous Knowledge (IK) across African institutions.	Institutional Alignment: Strong focus on aligning digitization with national heritage laws and library standards.	Bureaucracy: Heavy reliance on government institutions can slow down rapid, community-driven updates.	Aligns with the project's objective to partner with the NCCA and LGUs for legal sustainability and funding
Dive into Heritage	UNESCO (2024)	A digital platform utilizing 3D models and VR to document World Heritage sites and intangible practices.	Immersive Visualization: 3D and VR preservation offers the highest fidelity for tangible heritage (artifacts, buildings).	High Technical Barrier: Requires expensive hardware (LiDAR, VR headsets) not easily accessible to rural communities.	Future scalability option: The framework will be built to support 3D file formats for future scanning of Bagobo musical instruments/artifacts.
Traditional Knowledge Digital Library (TKDL)	Govt. of India (2001)	A defensive digital repository designed to protect traditional medicinal knowledge from patent piracy.	IP Protection: explicitly designed to prevent bio-piracy and unauthorized commercialization of indigenous knowledge.	Restricted Access: The defensive nature often limits educational access for the general public.	Security protocols from TKDL will be adapted to protect the "Indigenous Intellectual Property" of the Bagobo community.

Christen et al. (2017) developed Mukurtu CMS as an open-source content management system designed specifically for Indigenous communities to manage and protect digital cultural heritage. The system integrates cultural protocols that regulate access, sharing, and use of sensitive cultural materials based on community-defined rules. This study highlights that digital heritage systems can be effectively designed using Indigenous governance principles, ensuring that cultural ownership and control remain with the community while still enabling digitization and preservation of cultural resources.

Marayum

The University of the Philippines Diliman and DOST-ASTI developed Marayum Dictionary Platform as a collaborative online dictionary platform for Philippine Indigenous languages. The system enables crowdsourced contributions where native speakers can encode, validate, and refine language entries. This study demonstrates that community-driven digital platforms can play a significant role in preserving endangered languages by promoting participatory documentation and continuous language development.

Higaonon Mobile Dictionary

Researchers under MDPI (2025) developed Higaonon Mobile Dictionary as a mobile application aimed at preserving the Higaonon language in Mindanao. The application provides digitized vocabulary, pronunciation guides, and audio-based learning materials to support language retention and learning. This study shows that mobile-assisted

language learning systems can improve accessibility and contribute to the revitalization of endangered Indigenous languages, particularly among younger generations.

FirstVoices

The First Peoples' Cultural Council developed FirstVoices as a web-based platform for the documentation, preservation, and teaching of Indigenous languages. The system provides integrated tools such as dictionaries, keyboards, audio recordings, and storytelling archives to support both linguistic preservation and education. This study emphasizes that comprehensive digital language platforms are effective in combining archival preservation with interactive learning functionalities.

Irula Language MALL

The Cureus Journal (2025) presented Irula Language Mobile Assisted Language Learning App as a gamified mobile-assisted language learning application designed for the Irula tribal language. The system incorporates quizzes, rewards, and interactive exercises to enhance user engagement and learning retention. This study demonstrates that gamification techniques in mobile learning environments can significantly improve language acquisition, especially in the context of Indigenous and endangered languages.

Respectful Design Framework

Reitsma et al. (2013) proposed the Respectful Design Framework as a participatory approach to system development that prioritizes Indigenous ownership and involvement. The framework emphasizes co-design processes where communities actively participate in decision-making throughout system development. This study highlights the importance of ethical and culturally sensitive design in technology development for Indigenous knowledge systems.

WordWeaver

The National Research Council of Canada developed WordWeaver as an open-source linguistic tool that assists in the analysis and generation of complex verb structures in Indigenous languages. The system supports linguists and researchers in developing accurate language resources and documentation tools. This study shows that computational linguistic systems can enhance the structural analysis and preservation of endangered languages.

Digital Preservation of IK Framework

Biyela et al. (2016) developed the Digital Preservation of Indigenous Knowledge Framework as a structured model for digitizing Indigenous knowledge systems. The framework emphasizes ethical documentation, cultural context preservation, and community participation in all stages of digital archiving. This study highlights that

successful preservation of Indigenous knowledge requires balancing technological innovation with cultural integrity and participatory governance.

Dive Into Heritage

UNESCO (2024) developed Dive into Heritage as a digital platform that utilizes 3D modeling and virtual reality to document and present cultural heritage sites and intangible cultural practices. The system enables immersive exploration of heritage content for educational and preservation purposes. This study demonstrates that immersive technologies such as VR can enhance cultural education and improve engagement with heritage preservation initiatives.

Traditional Knowledge Digital Library (TKDL)

The Government of India established Traditional Knowledge Digital Library as a digital repository for documenting and protecting traditional medicinal knowledge systems such as Ayurveda, Unani, and Siddha. The system serves as a defensive mechanism against biopiracy by providing documented evidence of prior traditional knowledge. This study shows that digital knowledge repositories can effectively safeguard Indigenous and traditional knowledge from unauthorized patent claims.

The reviewed systems show a trend toward community-based, interactive, and ethical approaches in preserving language and culture. However, there is still a gap in fully integrating Indigenous dialects with AI-powered and culturally adaptive learning systems, which this study aims to address.

Conceptual Framework

The conceptual framework of the study illustrates the overall process involved in the development of the Web-Based Bagobo Tagabawa Cultural Preservation System. The framework begins with identifying the cultural and language preservation needs of the Bagobo Tagabawa community, particularly the risk of losing their dialect and cultural heritage due to modernization. To address these concerns, the system gathers cultural data through web platform inputs, audio and video recordings, document uploads, and community contributions.

The collected information is organized and stored in a centralized digital repository that serves as a secure database for cultural and language resources. To ensure authenticity and accuracy, the data undergo validation and monitoring through the participation of elders and cultural custodians. After validation, the information is processed by the AI Learning and Analytics Engine, which provides features such as AI chatbot support, language learning assistance, recommendation systems, and pattern analysis to improve user interaction and learning experiences.

The processed data are then utilized through the Decision Support module and the Web-Based Dashboard, which provide reports, monitoring tools, and cultural insights for users and administrators. The framework also emphasizes cultural implementation and community engagement through learning activities, cultural promotion, and language revitalization initiatives. Lastly, a feedback and continuous improvement loop ensures that user input and community updates are continuously integrated into the system to support the long-term preservation and sustainability of Bagobo Tagabawa culture and language.

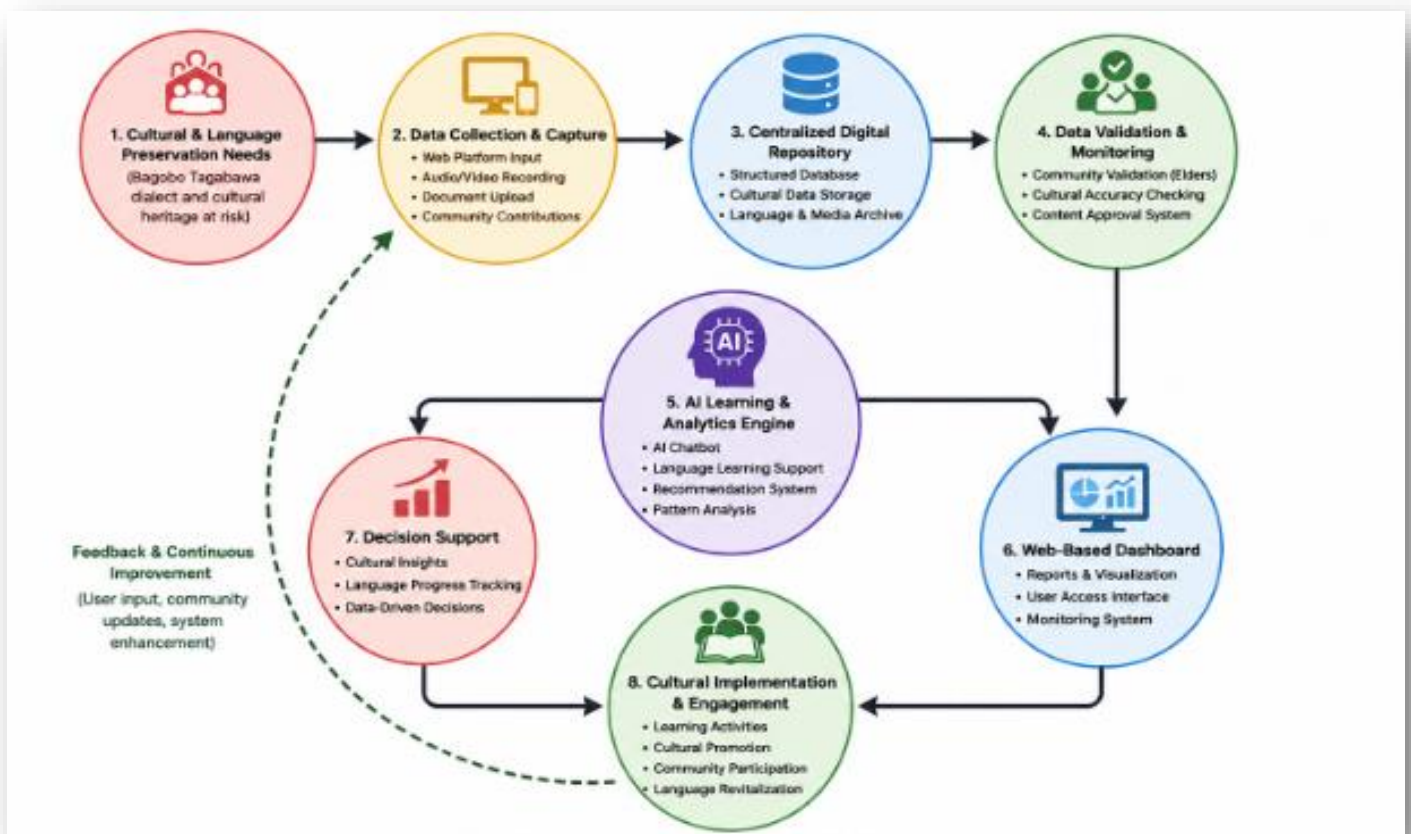


Figure 1: Conceptual Framework of the Study

CHAPTER III

METHODOLOGY

This chapter presents the methodology used in developing the proposed Web-Based Bagobo Tagabawa Cultural Preservation System. It discusses the research design, data gathering procedures, development process, and technologies used to ensure the effective preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Capstone Locale

The study will be conducted in the Municipality of Bansalan, Davao del Sur, Philippines as shown in the Figure 2 below. Bansalan is known for its rich cultural diversity and serves as one of the communities where the Bagobo Tagabawa people reside. The municipality possesses significant cultural resources, traditions, and linguistic heritage that contribute to the identity of the indigenous community.



Figure 2: Map of Bansalan

Bansalan is about 73 kilometers (45 mi) south of Davao City and is very accessible by land transportation mostly by buses going to Cotabato City, Kidapawan and Tacurong. Bansalan is about 17 kilometers (11 mi) north-west of the provincial capital city of Digos. The neighboring towns include Makilala in the north, Magsaysay in the west, Matanao in the south and Mount Apo and part of Digos in the east.

As modernization and globalization continue to influence local communities, there is an increasing need to preserve and promote the Bagobo Tagabawa dialect and cultural practices for future generations. The selection of Bansalan and some specific chosen Barangay's as the locale of the study is appropriate because of the presence of the Bagobo Tagabawa community and the availability of cultural knowledge that can support the development and implementation of the proposed Web-Based Bagobo Tagabawa Cultural Preservation System.

System Design

System design serves as the blueprint of the study, identifying the components, interfaces, and data flow necessary for the development of the proposed system. Figure 3 illustrates the design framework that provides the foundation of the study by organizing the components, interfaces, and data relationships needed in building the proposed system.

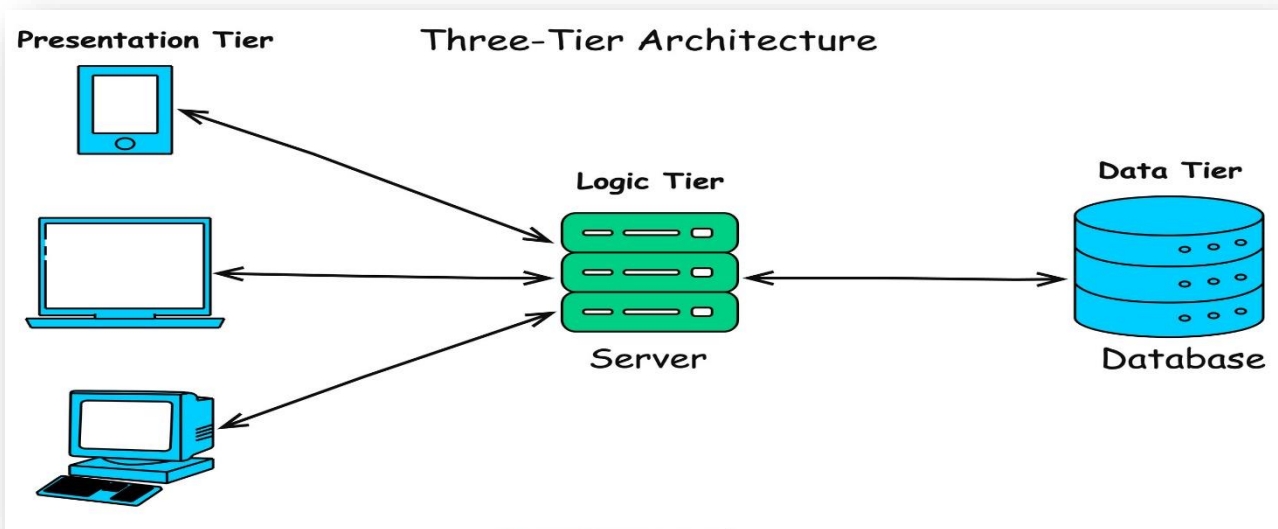


Figure 3: Three Tier Architecture Model of the Study

The three-tier architecture model was adopted in this study because it provides a structured and efficient framework for developing the proposed system. It divides the system into the presentation layer, business logic layer, and data layer, allowing each component to perform a distinct function. The presentation layer manages the user interface, the business logic layer handles the processing of system operations such as content management, learning activities, and AI-assisted functions, while the data layer is responsible for storing and retrieving the Bagobo Tagabawa cultural resources securely.

This architecture is suitable for the study because it promotes modularity, maintainability, and security. By separating the system into different layers, updates and enhancements can be implemented more easily without affecting the entire system. It also supports future expansion, making it appropriate for the long-term development of the proposed Bagobo Tagabawa cultural preservation platform.

Requirements

Requirements planning is the phase in the development process where the essential needs, functions, and constraints of the proposed system are identified, analyzed, and organized. It establishes the foundation for the entire project by determining what the system is expected to do, who will use it, and how it will support the objectives of the study. Through this stage, the researchers are able to define the scope of the system, clarify user requirements, and ensure that the proposed design is aligned with the actual needs of the target community.

In this study, requirements planning is important because it helps determine the features that must be included in the Bagobo Tagabawa cultural preservation system, such as the digital repository, learning modules, role-based access, and AI-assisted support. It also allows the researchers to identify the technical and operational limitations that may affect the development and implementation of the system. By carefully planning the requirements, the study can produce a system that is practical, relevant, and responsive to the needs of its intended users.

System Requirements

System requirements define the necessary hardware, software, functional capabilities, and operational conditions needed for the successful development and implementation of the proposed system. They serve as a guide in identifying the resources, technologies, and features required to ensure that the system performs

effectively, efficiently, and securely. Establishing clear system requirements helps developers understand the expected functionalities of the system and provides a basis for evaluating whether the system meets its intended objectives.

Software to be use. The development of the Epanaw Bagobo system will utilize various software tools and technologies to support the implementation of the web-based cultural preservation platform. The system will be developed using HTML, JavaScript, and Bootstrap for the front-end interface, while PHP with the Laravel Framework will be used for server-side processing and application logic. MySQL will serve as the database management system for storing user information, cultural resources, learning materials, and system records.

To support the AI-assisted learning component, an AI API service such as OpenAI or Google Gemini will be integrated into the system to provide chatbot functionality and learning assistance. The application will be hosted using Apache Web Server through XAMPP during development and may be deployed. Development and maintenance activities will be conducted using Visual Studio Code. Additional software such as Audacity, Canva, and Microsoft Office applications will be utilized for multimedia content preparation and project documentation.

Table 2: Software Recquirements of the Proposed System and its Specification

Software	Specification / Purpose
Windows 10/11	Operating System
Visual Studio Code	Code Editor
HTML, JavaScript	Front-End Development
PHP (Laravel)	Back-End Development
MySQL	Database Management System
OpenAI API / Gemini API	AI Chatbot Integration

XAMPP (Apache)	Local Server Environment
Google Chrome	Web Browser
Microsoft Office, Audacity, Canva	Documentation and Presentation

Hardware to be use. The hardware requirements of the proposed system consist of the physical devices needed to support the development, implementation, and operation of the Epanaw Bagobo system. A computer or laptop will be used as the primary device for system development, administration, and testing. A monitor will serve as the display output device, while a keyboard and mouse will be utilized for user input and system navigation. A microphone is required for recording Bagobo Tagabawa language pronunciations, interviews, and other audio-based cultural materials to be stored in the digital repository. An internet connection is necessary to enable access to the web-based platform and support AI chatbot integration. While, a learning material must also be needed as a basis for the Bagobo Tagabawa dialect. These hardware components are essential in supporting the functionality and sustainability of the proposed cultural preservation system.

Table 3: Hardware Recquirements of the Proposed System and its Specification

Hardware	Specification / Purpose
Computer/Laptop	System development and administration
Keyboard & Mouse	User input devices
Microphone, Phone Recorder	Audio recording for language materials
Internet Connection	System access and AI integration
Learning Materials	Serves as the basis.

Data Requirements

Data requirements refer to the necessary information that the system must collect, store, and manage in order to function properly. In this study, they include user data, cultural content, learning materials, AI interaction data, and administrative records that support the preservation and access of Bagobo Tagabawa dialect.

First Phase. For formality and acknowledgment, the researchers will ask for certification in the NCIP office regarding data collection and system making ahead of time.

Second Phase. The integration of the Epanaw Bagobo: An AI-Integrated Website Framework for the Preservation and Revitalization of the Bagobo Tagabawa Dialect and Cultural Heritage will be further explained after the confirmation of the NCIP office for the gathering of data.

Third Phase. The researchers identify the specific data required by the system, including cultural materials, language resources, user roles, learning content, and administrative information. This phase helped determine the scope of the data to be collected and ensured that the gathered information would support the objectives of the study.

Fourth Phase. The researchers gathered data through interviews, observations, and document review involving elders, community members, and other relevant respondents. The collected information included Bagobo Tagabawa words, stories, traditions, audio recordings, images, and other cultural references that would be used in the digital repository and learning modules of the system.

Fifth Phase. After the data were collected, they were reviewed and validated with the help of knowledgeable community members to ensure authenticity, accuracy, and cultural relevance. The researchers then organized the data according to category and purpose so that they could be properly stored and managed within the system.

Sixth Phase. The validated data were then used as inputs for the development of the proposed web-based system, particularly for the repository, learning module, and AI-assisted features. This phase ensured that the system would reflect the actual needs of the target users and would support the preservation and revitalization of Bagobo Tagabawa culture and language.

Data Dictionary

The data dictionary serves as a comprehensive reference that defines the data elements utilized in the proposed system. It provides a detailed description of each data field, including its field name, data type, field length, and purpose, to ensure consistency, accuracy, and integrity in the design and management of the system's database. Furthermore, it establishes standardized definitions for all data attributes, enabling efficient data storage, retrieval, and maintenance throughout the system development process.

For the proposed system, the data dictionary identifies the essential data required to support the system's core functionalities, including user management, digital repository

management, interactive learning modules, AI-assisted chatbot services, assessment records, and community contributions. It serves as a foundational document that guides the implementation of the database structure and ensures that all system data are organized, consistent, and aligned with the objectives of the study.

Dataset Table of the Project. The datasets sample in table's below simulate actual system data and ensure that the proposed system can effectively support user management, digital archiving, interactive learning, AI-assisted services, and community participation in preserving and promoting the Bagobo Tagabawa dialect and cultural heritage.

Table 4: Data Set Sample of User Account

User ID	Full Name	Email	Role
U001	Juan Dela Cruz	juan@gmail.com	Super Admin
U002	Maria Lumad	maria@gmail.com	Admin
U003	Carlo Ramos	carlo@gmail.com	User
U004	Ana Torres	ana@gmail.com	User

Table 4 presents the sample records of individuals who are authorized to access the proposed system. It includes essential user information such as the User ID, Full Name, Email Address, Role, and Account Status. The User ID serves as the unique identifier for each registered user, while the Full Name and Email Address are used for user identification and authentication. The Role field specifies whether the user is a Super

Admin, Admin, or User, which determines the level of access and permissions granted within the system. The Account Status indicates whether the user's account is active or inactive. This dataset demonstrates the implementation of the Role-Based Access Control (RBAC) mechanism, ensuring that system functions and resources are accessed only by authorized users according to their assigned responsibilities.

Table 5: Data Set Sample of Cultural Repository

Resource ID	Title	Category	Description	Uploader
CR001	Bagobo Traditional Dance	Video	Demonstration of traditional Bagobo dance.	Admin
CR002	Bagobo Greeting Words	Audio	Audio pronunciation of common greetings.	Admin
CR003	Bagobo Traditional Attire	Image	Traditional clothing worn during rituals.	Admin
CR004	The Legend of Mt. Apo	Text	A traditional Bagobo folktale.	Admin

The table 5 presents the sample dataset of the digital cultural resources stored within the proposed system. It includes the Resource ID, Title, Category, Description, and Uploaded By fields. Each Resource ID uniquely identifies a cultural resource, while the Title provides the name of the resource. The Category classifies the resource into audio, video, image, or text, allowing users to easily organize and locate cultural materials. The Description provides a brief explanation of the content, and the Uploaded By field identifies the administrator responsible for uploading the resource. This dataset demonstrates how the system stores, organizes, and manages Bagobo Tagabawa cultural

materials in a centralized digital repository, supporting the preservation and accessibility of indigenous language and cultural heritage.

Table 6: Data Set Sample of Learning Materials

Lesson ID	Lesson Title	Module	Difficulty
L001	Basic Bagobo Greetings	Language	Beginner
L002	Numbers in Bagobo	Language	Beginner
L003	Bagobo Cultural Practices	Culture	Intermediate
L004	Traditional Festivals	Culture	Intermediate

The table 6 contains the sample educational content that users can access through the interactive learning module of the proposed system. It includes the Lesson ID, Lesson Title, Module, and Difficulty fields. The Lesson ID uniquely identifies each lesson, while the Lesson Title describes the topic being discussed. The Module field categorizes the lesson into specific learning areas, such as Language or Culture, and the Difficulty field identifies the complexity level of each lesson to guide learners according to their knowledge and learning progress. This dataset demonstrates how the proposed system delivers structured educational content designed to promote the learning and revitalization of the Bagobo Tagabawa dialect, traditions, and cultural practices.

Table 7: Data Set Sample of Quiz Results

Result ID	User	Lesson	Score	Remarks
Q001	Carlo Ramos	Basic Bagobo Greetings	95	Passed
Q002	Ana Torres	Numbers in Bagobo	88	Passed
Q003	Carlo Ramos	Cultural Practices	90	Passed

The table 7 presents the sample assessment records generated after users complete the learning activities provided by the system. It contains the Result ID, User, Lesson, Score, and Remarks fields. The Result ID uniquely identifies each assessment record, while the User field identifies the learner who completed the quiz. The Lesson field specifies the lesson associated with the assessment, and the Score records the user's performance. The Remarks field indicates whether the learner has successfully passed the assessment based on the established criteria. This dataset demonstrates how the system evaluates user learning outcomes, monitors academic progress, and provides performance records that can be used to measure the effectiveness of the interactive learning module.

Table 8: Data Set Sample of Chatbot Logs

Chat ID	User	Question	AI Response
C001	Carlo Ramos	How do you say "Thank you" in Bagobo?	Displays the Bagobo translation and pronunciation.
C002	Ana Torres	Tell me about Bagobo traditions.	Provides an overview of Bagobo cultural traditions.

The table 8 contains the sample records of interactions between users and the AI-powered chatbot integrated into the proposed system. It includes the Chat ID, User, Question, and AI Response fields. The Chat ID uniquely identifies each chatbot interaction, while the User field identifies the individual who submitted the inquiry. The Question field records the user's request or inquiry related to the Bagobo Tagabawa

dialect or cultural heritage, and the AI Response field stores the corresponding answer generated by the chatbot. This dataset demonstrates how the AI-assisted learning component provides immediate responses, language assistance, and cultural information, thereby enhancing user engagement and supporting self-paced learning within the proposed web-based platform.

Table 9: Data Set Sample of Community Contributions

Contribution ID	Contributor	Title	Type	Status
CC001	Maria Lumad	Traditional Wedding Ritual	Text	Approved
CC002	Ana Torres	Bagobo Harvest Festival	Image	Pending
CC003	Carlo Ramos	Bagobo Folk Song	Audio	Approved

The table 9 presents the sample records of cultural content submitted by members of the Bagobo Tagabawa community through the proposed system. It includes the Contribution ID, Contributor, Title, Type, and Status fields. The Contribution ID uniquely identifies each submitted resource, while the Contributor field records the individual who submitted the content. The Title identifies the cultural material, and the Type specifies whether the submission is in the form of text, image, audio, or another supported media type. The Status field indicates whether the submitted content is Pending or Approved after undergoing administrative review and validation. This dataset demonstrates the community participation feature of the proposed system by allowing users to contribute valuable cultural knowledge while ensuring that all submissions are reviewed for authenticity, relevance, and accuracy before being included in the digital repository. This

process supports the collaborative preservation and continuous enrichment of the Bagobo Tagabawa dialect and cultural heritage.

Table 10: Data Set Sample of Users Feedback

Contribution ID	Contributor	Title	Type	Status
CC001	Maria Lumad	Traditional Wedding Ritual	Text	Approved
CC002	Ana Torres	Bagobo Harvest Festival	Image	Pending
CC003	Carlo Ramos	Bagobo Folk Song	Audio	Approved

The table 10 contains the sample records of evaluations and comments submitted by users regarding their experience with the proposed system. It includes the Feedback ID, User, Rating, and Comments fields. The Feedback ID serves as the unique identifier for each feedback entry, while the User field identifies the individual who submitted the evaluation. The Rating field allows users to assess the system's overall performance, usability, and effectiveness using a numerical scale, whereas the Comments field provides qualitative feedback, suggestions, or observations about the system's features and functionality. This dataset demonstrates how the proposed system collects user evaluations to measure user satisfaction, identify areas for improvement, and support the continuous enhancement of the web-based platform. Furthermore, the collected feedback serves as valuable input for assessing the system's effectiveness in preserving, promoting, and revitalizing the Bagobo Tagabawa dialect and cultural heritage while improving the overall user experience.

Table 11: Data Set Sample of Vocabulary Dictionary

Word ID	Bagobo Word	English Meaning	Pronunciation	Category
W001	Genda	Where	Ge-n-da	Greeting
W002	Salamat	Thank You	sa-la-mat	Expression
W003	Wayig	Water	Wa-y-ig	Noun
W004	Apo	Grandparent	a-po	Family

The table 11 contains the sample records of Bagobo Tagabawa words and their corresponding English translations. It includes the Word ID, Bagobo Word, English Meaning, Pronunciation, and Category fields. The Word ID serves as the unique identifier for each vocabulary entry, while the Bagobo Word stores the indigenous term and the English Meaning provides its corresponding translation. The Pronunciation field serves as a guide for users to correctly pronounce the word, and the Category field classifies the vocabulary according to its type, such as greetings, expressions, nouns, verbs, or adjectives. This dataset demonstrates how the proposed system preserves and documents the Bagobo Tagabawa dialect while providing users with structured language resources that support vocabulary development and language revitalization.

Table 12: Data Set Sample of Learning Progress

Progress ID	User	Lesson	Status	Date Completed
P001	Carlo Ramos	Basic Bagobo Greetings	Completed	March 15, 2026
P002	Ana Torres	Numbers in Bagobo	In Progress	—
P003	Carlo Ramos	Cultural Practices	Completed	March 18, 2026
P004	Ana Torres	Traditional Festivals	Not Started	—

The table 12 contains the sample records of users' progress throughout the learning modules provided by the proposed system. It includes the Progress ID, User, Lesson, Status, and Date Completed fields. The Progress ID uniquely identifies each learning record, while the User field identifies the learner whose progress is being monitored. The Lesson field specifies the learning module being undertaken, and the Status field indicates whether the lesson is Completed, In Progress, or Not Started. The Date Completed field records the date on which the learner successfully completed the lesson. This dataset demonstrates how the proposed system monitors learner participation and tracks learning achievements, allowing users to continue their lessons efficiently while enabling administrators to monitor overall learning performance.

Table 13: Data Set Sample of Cultural Events

Event ID	Event Name	Date	Location	Description
E001	Harvest Festival	August 15	Bansalan	Annual thanksgiving celebration.
E002	Cultural Dance Festival	October 2	Davao del Sur	Traditional Bagobo dances.
E003	Storytelling Session	July 20	Barangay Kinuskusan	Sharing of Bagobo folktales.

The table 13 contains the sample records of traditional Bagobo Tagabawa cultural activities and celebrations documented in the proposed system. It includes the Event ID, Event Name, Date, Location, and Description fields. The Event ID uniquely identifies each cultural event, while the Event Name specifies the title of the activity. The Date records the scheduled or commemorative date of the event, and the Location identifies where

the activity takes place. The Description field provides a brief explanation of the event and its cultural significance. This dataset demonstrates how the proposed system documents and promotes traditional ceremonies, festivals, and cultural activities to encourage awareness, participation, and appreciation of the Bagobo Tagabawa cultural heritage.

Table 14: Data Set Sample of Activity Logs

Log ID	User	Activity	Date
A001	Maria Lumad	Uploaded new audio resource	March 10, 2026
A002	Juan Dela Cruz	Approved community contribution	March 11, 2026
A003	Maria Lumad	Updated learning lesson	March 15, 2026
A004	Juan Dela Cruz	Created new user account	March 18, 2026

The table 14 contains the sample records of significant actions performed by users and administrators within the proposed system. It includes the Log ID, User, Activity, and Date fields. The Log ID uniquely identifies each recorded activity, while the User field identifies the individual who performed the action. The Activity field describes the specific task completed, such as uploading cultural resources, approving community contributions, updating learning materials, or managing user accounts. The Date field records when the activity occurred. This dataset demonstrates how the proposed system maintains an audit trail of system activities to support monitoring, accountability, system maintenance, and administrative reporting.

Table 15: Data Set Sample of AI Learning Recommendations

Recommendation ID	User	Recommended Lesson	Reason
R001	Carlo Ramos	Numbers in Bagobo	Beginner-level lesson recommended.
R002	Ana Torres	Traditional Festivals	Based on previous lesson completion.
R003	Carlo Ramos	Bagobo Folk Songs	Suggested after completing cultural lessons.

The table 15 contains the sample records of lesson recommendations generated by the AI component of the proposed system. It includes the Recommendation ID, User, Recommended Lesson, and Reason fields. The Recommendation ID uniquely identifies each recommendation, while the User field specifies the learner who receives the recommendation. The Recommended Lesson identifies the lesson suggested by the AI, and the Reason field explains why the lesson was recommended, such as based on quiz performance, learning progress, or previous activities. This dataset demonstrates the adaptive learning capability of the proposed system by providing personalized learning recommendations that help users improve their understanding of the Bagobo Tagabawa dialect and cultural heritage.

Table 16: Data Set Sample of Cultural Practices

Practice ID	Practice Name	Category	Description
CP001	Gin-Em Ceremony	Ritual	Traditional thanksgiving ceremony.
CP002	Harvest Festival	Festival	Celebration after a successful harvest.
CP003	Traditional Courtship	Tradition	Indigenous courtship practices.

CP004	Wedding Ceremony	Ceremony	Traditional Bagobo marriage customs.
CP005	Tribal Council Meeting	Governance	Gathering led by tribal elders.
CP006	Healing Ritual	Ritual	Traditional healing practices using indigenous knowledge.

The table 16 contains sample records of the customs, traditions, ceremonies, and rituals practiced by the Bagobo Tagabawa community. It includes the Practice ID, Practice Name, Category, and Description fields. The Practice ID uniquely identifies each cultural practice, while the Practice Name specifies its title. The Category classifies the practice according to its purpose, such as rituals, festivals, or ceremonies, and the Description provides a brief explanation of its cultural significance. This dataset supports the documentation and preservation of indigenous customs and traditions through organized digital records.

Table 17: Data Set Sample of Storytelling Archive

Story ID	Story Title	Storyteller	Category	Language
ST001	Legend of Mt. Apo	Datu Ramon	Folktale	Bagobo
ST002	The Brave Hunter	Elder Juan	Legend	Bagobo
ST003	The Sacred Tree	Elder Rosa	Myth	Bagobo
ST004	Origin of the River	Datu Ramon	Folktale	Bagobo
ST005	The Spirit of the Forest	Elder Lina	Legend	Bagobo
ST006	The First Bagobo Family	Elder Pedro	Oral Tradition	Bagobo

The table 17 contains sample records of traditional Bagobo Tagabawa stories documented in the proposed system. It includes the Story ID, Story Title, Storyteller, Category, and Language fields. The Story ID uniquely identifies each story, while the Story Title provides the title of the cultural narrative. The Storyteller identifies the

individual who shared the story, the Category classifies the story according to its type, and the Language specifies the language used in the narration. This dataset demonstrates how the proposed system digitally preserves oral traditions, legends, myths, and folktales for future generations.

Table 18: Data Set Sample of Pronunciation Library

Media ID	Title	Media Type	Category	Uploaded By	Date Uploaded
M001	Traditional Bagobo Dance	Video	Cultural Dance	Admin Maria	March 10, 2026
M002	Bagobo Greeting Pronunciation	Audio	Language	Admin Pedro	March 11, 2026
M003	Traditional Bagobo Attire	Image	Cultural Attire	Admin Maria	March 12, 2026
M004	Harvest Festival Celebration	Video	Festival	Admin Pedro	March 14, 2026
M005	Indigenous Musical Instruments	Image	Cultural Artifacts	Admin Maria	March 15, 2026
M006	Traditional Folk Song	Audio	Music	Admin Pedro	March 16, 2026
M007	Bagobo Wedding Ceremony	Video	Tradition	Admin Maria	March 18, 2026
M008	Traditional Weaving Patterns	Image	Handicrafts	Admin Pedro	March 20, 2026

The table 18 contains the sample records of Bagobo Tagabawa vocabulary with their corresponding pronunciation resources. It includes the Pronunciation ID, Bagobo Word, English Meaning, Audio File, Native Speaker, and Verified By fields. The Pronunciation ID serves as the unique identifier for each pronunciation record, while the Bagobo Word and English Meaning provide the indigenous word and its English translation. The Audio File stores the recorded pronunciation spoken by a native speaker,

the Native Speaker field identifies the individual who provided the recording, and the Verified By field records the administrator or cultural elder who validated the pronunciation for accuracy and authenticity. This dataset demonstrates how the proposed system preserves the correct pronunciation of the Bagobo Tagabawa dialect and supports the interactive learning module and AI-assisted chatbot by providing verified audio references for language learners.

Table 19: Data Set Sample of Multimedia Gallery

Media ID	Title	Media Type	Category	Uploaded By	Date Uploaded
M001	Traditional Bagobo Dance	Video	Cultural Dance	Admin Maria	March 10, 2026
M002	Bagobo Greeting Pronunciation	Audio	Language	Admin Pedro	March 11, 2026
M003	Traditional Bagobo Attire	Image	Cultural Attire	Admin Maria	March 12, 2026
M004	Harvest Festival Celebration	Video	Festival	Admin Pedro	March 14, 2026
M005	Indigenous Musical Instruments	Image	Cultural Artifacts	Admin Maria	March 15, 2026
M006	Traditional Folk Song	Audio	Music	Admin Pedro	March 16, 2026
M007	Bagobo Wedding Ceremony	Video	Tradition	Admin Maria	March 18, 2026
M008	Traditional Weaving Patterns	Image	Handicrafts	Admin Pedro	March 20, 2026

The table 19 contains the sample records of multimedia resources stored in the proposed system to support the preservation and promotion of the Bagobo Tagabawa dialect and cultural heritage. It includes the Media ID, Title, Media Type, Category, Uploaded By, and Date Uploaded fields. The Media ID serves as the unique identifier for

each multimedia resource, while the Title specifies the name of the media content. The Media Type identifies the format of the resource, such as image, audio, or video, and the Category classifies the content based on its cultural significance, including language, traditions, festivals, rituals, and cultural artifacts. The Uploaded By field records the administrator responsible for uploading the resource, while the Date Uploaded indicates when the multimedia file was added to the system. This dataset demonstrates how the proposed system organizes and manages digital multimedia resources, providing users with accessible and engaging visual and audio materials that support cultural preservation, language learning, and community awareness.

Table 20: Data Set Sample of Resource Verification

Verification ID	Resource Title	Verified By	Date Verified	Status
V001	Bagobo Folk Song	Datu Ramon	March 10, 2026	Verified
V002	Traditional Dance	Elder Juan	March 12, 2026	Verified
V003	Bagobo Greeting Words	Elder Rosa	March 14, 2026	Verified
V004	Legend of Mt. Apo	Datu Ramon	March 18, 2026	Verified
V005	Traditional Attire	Elder Lina	March 20, 2026	Pending
V006	Healing Ritual	Elder Pedro	March 22, 2026	Verified

The table 20 contains sample records of cultural resources that have undergone validation by cultural elders or authorized administrators. It includes the Verification ID, Resource Title, Verified By, Date Verified, and Status fields. The Verification ID uniquely identifies each verification record, while the Resource Title identifies the cultural material being reviewed. The Verified By field records the cultural elder or administrator responsible for the validation process, the Date Verified indicates when the verification

was completed, and the Status field specifies whether the resource has been approved or is pending review. This dataset demonstrates the proposed system's commitment to maintaining the authenticity, accuracy, and credibility of the Bagobo Tagabawa cultural resources before they are published in the digital repository.

User Design

The user design for Epanaw Bagobo is structured around a three-tier architecture that bridges the digital gap for the preservation of the Bagobo Tagabawa dialect and culture in Bansalan, Davao del Sur.

Diagrams

Diagrams are essential tools and serve as visual aids that simplify and clarify complex information. Diagrams effectively illustrate relationships, processes, and data that would be cumbersome or difficult to explain through text alone. By breaking down intricate concepts into digestible visuals, diagrams enhance readability, make your work more engaging, and help readers quickly grasp key ideas. Ultimately, they strengthen your arguments, provide clear evidence for your findings, and improve the overall professional quality and impact of your capstone project.

Entity Relationship Diagram. The Entity Relationship Diagram (ERD) illustrates the logical structure of the database for the proposed capstone. It identifies the system's primary entities, their attributes, and the relationships among them. The ERD serves as the blueprint for the database design by defining how data are organized, stored, and interconnected to support the system's functionalities. It also ensures data consistency, minimizes redundancy, and facilitates efficient data retrieval and management. As shown in the ERD, the User Accounts entity serves as the central table of the database since it manages the information of all registered users, including Super Administrators, Administrators, and Users. Each user can access the system according to the permissions assigned through the Role-Based Access Control (RBAC) mechanism. The User Accounts entity is related to several modules, including Learning Progress, Quiz Results, AI Chatbot Logs, Feedback, Community Contributions, and Activity Logs, allowing the system to monitor user activities, learning records, chatbot interactions, submitted contributions, and system usage. The Learning Materials entity stores the educational content provided by the system, including language lessons and cultural learning modules. Each learning material can be accessed by multiple users, while each user can complete several lessons. The Learning Progress and Quiz Results entities are associated with both User Accounts and Learning Materials to monitor lesson completion, assessment scores, and overall learning performance. The Vocabulary Dictionary entity contains the Bagobo Tagabawa vocabulary entries together with their English translations and learning information. Each vocabulary entry is associated with the Pronunciation Library, which stores verified audio recordings of native speakers. This relationship allows

users to learn not only the meaning of indigenous words but also their correct pronunciation, thereby supporting the revitalization of the Bagobo Tagabawa dialect. The Cultural Repository entity serves as the primary storage for digital cultural resources, including documents, images, audio recordings, and videos related to the Bagobo Tagabawa community. This repository is connected to the Multimedia Gallery, which organizes multimedia files for easier browsing and presentation. Likewise, the repository is linked to the Storytelling Archive, Cultural Practices, and Cultural Events entities, enabling the system to preserve traditional stories, customs, rituals, ceremonies, festivals, and other significant cultural information in a structured manner. To maintain the authenticity and integrity of cultural resources, the Community Contributions entity records the cultural materials submitted by registered users, while the Resource Verification entity stores the validation records performed by administrators or recognized cultural custodians. This process ensures that all published cultural resources have undergone proper review before becoming accessible to other users. The AI Chatbot Logs entity records user interactions with the AI-powered chatbot by storing users' questions and the corresponding responses generated by the system. These records support system monitoring and provide insights that can be used to improve the chatbot's performance. Similarly, the Feedback entity collects users' ratings, comments, and suggestions regarding the usability and effectiveness of the system. These evaluations provide valuable information for the continuous enhancement of the platform. Finally, the Activity Logs entity records significant actions performed by users and administrators, such as uploading cultural resources, approving contributions, updating learning materials, and

managing user accounts. This entity supports administrative monitoring, accountability, and system auditing by maintaining a complete record of important system activities. Overall, the Entity Relationship Diagram demonstrates the relationships among the core entities required to support the proposed system. It provides a well-structured database design that enables efficient management of user information, digital cultural resources, language learning modules, AI-assisted services, multimedia content, community participation, and administrative functions. Through these interconnected entities, the proposed system effectively supports its primary objective of preserving and revitalizing the Bagobo Tagabawa dialect and cultural heritage using a secure, organized, and AI-integrated web-based platform.

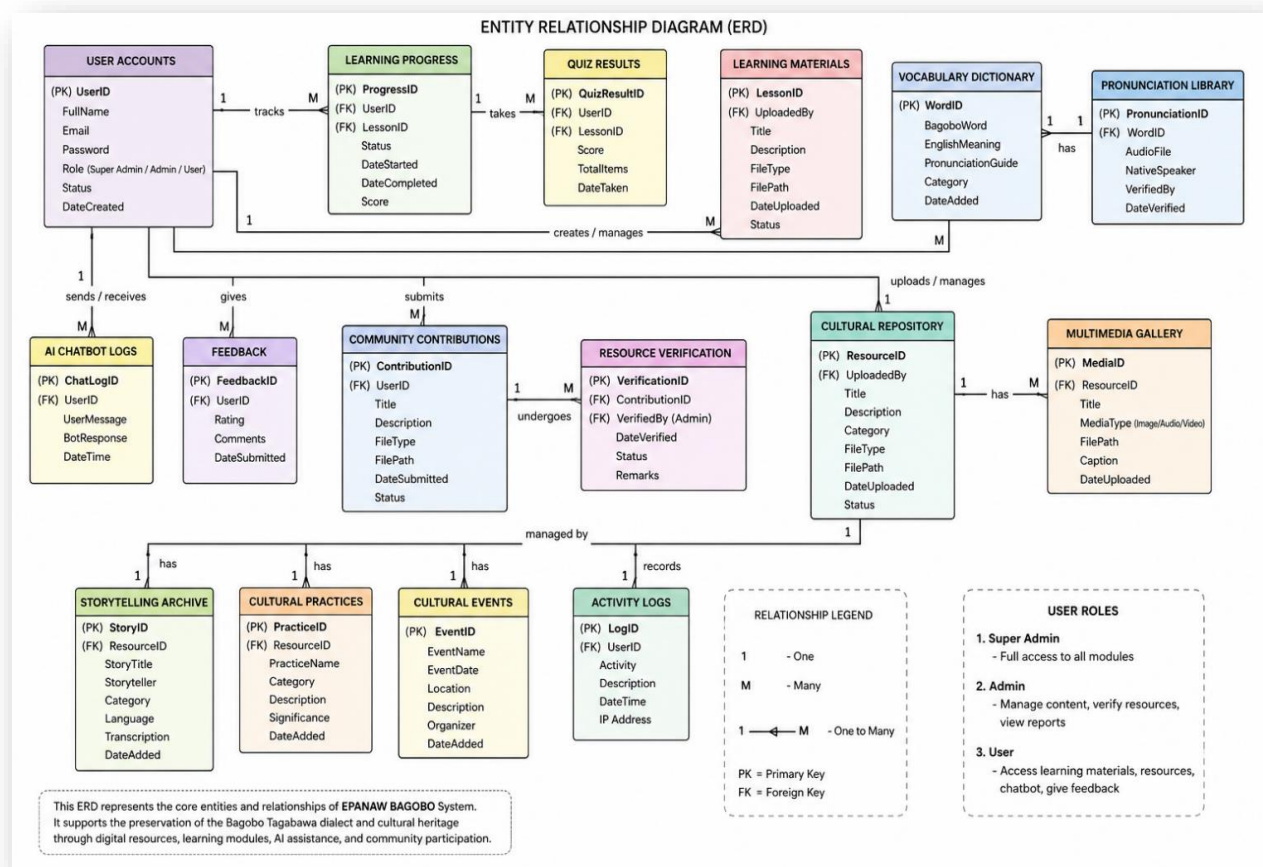


Figure 5: Entity Relationship Diagram of the Study

Database Schema. The Database Schema presents the logical structure of the database used in the proposed system. It defines the database tables, their corresponding fields, data types, primary keys, and foreign keys that support the system's functionalities. The schema serves as the foundation for database implementation by ensuring data consistency, integrity, and efficient management of user information, cultural resources, learning materials, AI-assisted services, and community contributions.

Table 21: Database Schema of User Account

Field Name	Data Type	Constraints
UserID	INT	Primary Key, Auto Increment
FullName	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	UNIQUE, NOT NULL
Password	VARCHAR(255)	NOT NULL
Role	ENUM('Super Admin','Admin','User')	NOT NULL
Status	VARCHAR(20)	NOT NULL
DateCreated	DATETIME	NOT NULL

The table 21 stores the information of all registered users in the proposed system. It contains user credentials and account details, including the assigned role and account status. This table supports the implementation of the Role-Based Access Control (RBAC) mechanism by defining the access privileges of the Super Administrator, Administrator, and User. It also serves as the primary reference for user authentication and system access management.

Table 22: Database Schema of Learning Materials

Field Name	Data Type	Constraints
LessonID	INT	Primary Key, Auto Increment
Title	VARCHAR(150)	NOT NULL

Description	TEXT	NOT NULL
Module	VARCHAR(50)	NOT NULL
Difficulty	VARCHAR(30)	NOT NULL
UploadedBy	INT	Foreign Key (UserID)

The table 22 contains the educational content provided by the proposed system. It stores information about language lessons, cultural topics, module categories, and difficulty levels. This table serves as the primary source of learning resources that enable users to study the Bagobo Tagabawa dialect and cultural heritage through structured and interactive lessons.

Table 23: Database Schema of Learning Progress

Field Name	Data Type	Constraints
ProgressID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key
LessonID	INT	Foreign Key
Status	VARCHAR(30)	NOT NULL
DateCompleted	DATE	NULL

The table 23 records the progress of each user throughout the learning modules. It stores information regarding lesson completion status and completion dates, allowing the system to monitor users' learning achievements. This table enables learners to track their progress while providing administrators with data for evaluating learning participation.

Table 24: Database Schema of Quiz Results

Field Name	Data Type	Constraints
QuizResultID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key

LessonID	INT	Foreign Key
Score	INT	NOT NULL
Remarks	VARCHAR(20)	NOT NULL

The table 24 stores the assessment records generated after user's complete quizzes associated with the learning modules. It contains users' scores and evaluation remarks, allowing the system to measure learning outcomes and monitor users' academic performance. This table supports the assessment component of the interactive learning module.

Table 25: Data Schema of Vocabulary Dictionary

Field Name	Data Type	Constraints
WordID	INT	Primary Key, Auto Increment
BagoboWord	VARCHAR(100)	NOT NULL
EnglishMeaning	VARCHAR(100)	NOT NULL
Pronunciation	VARCHAR(100)	NOT NULL
Category	VARCHAR(50)	NOT NULL

The table 25 contains the Bagobo Tagabawa vocabulary entries used within the proposed system. It stores indigenous words together with their English translations, pronunciation guides, and language categories. This table supports language learning by providing users with a structured digital dictionary that promotes vocabulary development and language preservation.

Table 26: Data Schema of Pronunciation Library

Field Name	Data Type	Constraints
PronunciationID	INT	Primary Key, Auto Increment
WordID	INT	Foreign Key
AudioFile	VARCHAR(255)	NOT NULL

NativeSpeaker	VARCHAR(100)	NOT NULL
VerifiedBy	INT	Foreign Key (UserID)

The table 26 stores the verified pronunciation records of Bagobo Tagabawa vocabulary. It contains audio files, native speaker information, and verification details to ensure pronunciation accuracy. This table supports the interactive learning module and AI-assisted chatbot by providing authentic pronunciation references for users learning the indigenous dialect.

Table 27: Data Schema of Cultural Repository

Field Name	Data Type	Constraints
ResourceID	INT	Primary Key, Auto Increment
UploadedBy	INT	Foreign Key
Title	VARCHAR(150)	NOT NULL
Description	TEXT	NOT NULL
Category	VARCHAR(50)	NOT NULL
FileType	VARCHAR(30)	NOT NULL
FilePath	VARCHAR(255)	NOT NULL
DateUploaded	DATETIME	NOT NULL

The table 27 serves as the central repository for the digital cultural resources stored in the proposed system. It contains information about uploaded documents, images, audio recordings, videos, and other cultural materials related to the Bagobo Tagabawa community. This table supports the preservation, organization, and retrieval of indigenous cultural heritage in a centralized database.

Table 28: Data Schema of Multimedia Gallery

Field Name	Data Type	Constraints
MediaID	INT	Primary Key, Auto Increment
ResourceID	INT	Foreign Key
Title	VARCHAR(150)	NOT NULL

MediaType	VARCHAR(30)	NOT NULL
FilePath	VARCHAR(255)	NOT NULL
Caption	TEXT	NULL

The table 28 stores multimedia resources associated with the cultural repository. It contains images, audio recordings, and videos that enhance users' learning experience by providing visual and audio representations of Bagobo Tagabawa traditions, language, ceremonies, and cultural artifacts. This table improves the accessibility and presentation of digital cultural resources.

Table 29: Data Schema of Storytelling Archive

Field Name	Data Type	Constraints
StoryID	INT	Primary Key, Auto Increment
ResourceID	INT	Foreign Key
StoryTitle	VARCHAR(150)	NOT NULL
Storyteller	VARCHAR(100)	NOT NULL
Category	VARCHAR(50)	NOT NULL
Language	VARCHAR(50)	NOT NULL

The table 29 contains digital records of traditional Bagobo Tagabawa stories, legends, myths, and oral traditions. It stores information regarding the story title, storyteller, category, and language used. This table contributes to the preservation of indigenous oral literature and supports cultural education through digital storytelling.

Table 30: Data Schema of Cultural Practices

Field Name	Data Type	Constraints
PracticeID	INT	Primary Key, Auto Increment
ResourceID	INT	Foreign Key
PracticeName	VARCHAR(150)	NOT NULL
Category	VARCHAR(50)	NOT NULL
Description	TEXT	NOT NULL

The table 30 stores information about the customs, traditions, rituals, ceremonies, and cultural activities of the Bagobo Tagabawa community. It provides descriptions and classifications of each cultural practice, allowing users to access documented information that promotes cultural awareness and heritage preservation.

Table 31: Data Schema of Community Contributions

Field Name	Data Type	Constraints
ContributionID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key
Title	VARCHAR(150)	NOT NULL
Description	TEXT	NOT NULL
FileType	VARCHAR(30)	NOT NULL
FilePath	VARCHAR(255)	NOT NULL
Status	VARCHAR(20)	NOT NULL

The table 31 records cultural materials submitted by registered users of the proposed system. It stores information about user-submitted documents, images, audio recordings, and other resources, including their approval status. This table encourages community participation while allowing administrators to review and manage contributed cultural content before publication.

Table 32: Data Schema of Resource Verification

Field Name	Data Type	Constraints
VerificationID	INT	Primary Key, Auto Increment
ContributionID	INT	Foreign Key
VerifiedBy	INT	Foreign Key (UserID)
DateVerified	DATE	NOT NULL
Status	VARCHAR(20)	NOT NULL
Remarks	TEXT	NULL

The table 32 stores the validation records of cultural resources submitted to the proposed system. It records the administrator or cultural elder responsible for verifying each resource, together with the verification date and status. This table ensures the authenticity, reliability, and cultural accuracy of the information published in the digital repository.

Table 33: Data Schema of AI Chatbot Logs

Field Name	Data Type	Constraints
ChatLogID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key
UserMessage	TEXT	NOT NULL
BotResponse	TEXT	NOT NULL
DateTime	DATETIME	NOT NULL

The table 33 records the interactions between users and the AI-powered chatbot. It stores users' inquiries together with the responses generated by the AI assistant and the corresponding date and time of each interaction. This table supports system monitoring, evaluates chatbot performance, and helps improve AI-assisted learning services.

Table 34: Data Schema of Feedback

Field Name	Data Type	Constraints
FeedbackID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key
Rating	INT	NOT NULL
Comments	TEXT	NULL
DateSubmitted	DATETIME	NOT NULL

The table 34 stores the evaluations, comments, and suggestions submitted by users regarding the proposed system. It records users' ratings together with their feedback to assess system usability, functionality, and overall user satisfaction. This table provides valuable information that supports continuous system improvement and enhances the quality of the learning platform.

Table 35: Data Schema of Activity Logs

Field Name	Data Type	Constraints
LogID	INT	Primary Key, Auto Increment
UserID	INT	Foreign Key
Activity	VARCHAR (255)	NOT NULL
Description	TEXT	NULL
DateTime	DATETIME	NOT NULL

The table 35 records significant activities performed by users and administrators within the proposed system. It stores information about user actions, activity descriptions, and timestamps to maintain an audit trail of system operations. This table supports system monitoring, administrative reporting, accountability, and security by tracking important events occurring within the application.

Use Case Diagram. The Use Case Diagram illustrates the functional requirements of the system. It identifies the different users of the system and the specific functions they can perform based on their assigned roles. The diagram provides a visual representation of the interactions between the system and its actors, ensuring that the system's features are aligned with the objectives and requirements of the study. As illustrated in the Use Case Diagram shows in Figure 6 below , the proposed system

consists of three primary actors: the Super Admin, Admin, and User. Each actor is assigned specific roles and permissions through the implementation of Role-Based Access Control (RBAC), ensuring secure and authorized access to the system's functionalities. The Super Admin has the highest level of access and is responsible for the overall management and administration of the system. This actor manages user and administrator accounts, oversees learning materials and cultural resources, maintains the multimedia gallery, vocabulary dictionary, pronunciation library, storytelling archive, and cultural practices, verifies community contributions, manages cultural events, monitors user feedback and activity logs, generates system reports, and ensures the efficient operation of the platform. The admin is responsible for managing the system's educational and cultural content. The admin can upload and update learning materials, cultural resources, multimedia content, vocabulary entries, pronunciation records, storytelling archives, and cultural practices. In addition, the admin reviews community contributions, manages cultural events, monitors user feedback, and generates reports to support the maintenance and continuous improvement of the system. The User represents the learners and community members who utilize the system for educational and cultural purposes. Registered users can create and manage their accounts, browse the digital cultural repository, access multimedia resources, search vocabulary entries, listen to pronunciation guides, read traditional stories, explore cultural practices and events, access learning modules, complete quizzes, monitor their learning progress, interact with the AI-powered chatbot, submit community contributions, and provide feedback regarding the system. Furthermore, the diagram illustrates the relationships

among the system's use cases through «include» relationships, indicating that certain processes are dependent on or integrated with other system functions. For example, user authentication is required before accessing protected features, while community contributions undergo a verification process before becoming available in the digital repository. Similarly, report generation and content management involve multiple interconnected processes that support the overall functionality of the proposed system. Overall, the Use Case Diagram demonstrates the interactions between the system and its intended users by illustrating the major functional requirements of the proposed platform. It provides a clear representation of how the system facilitates user management, digital cultural preservation, interactive learning, AI-assisted services, and community participation. These interactions ensure that the proposed system effectively supports the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage while providing a secure, organized, and user-centered web-based platform.

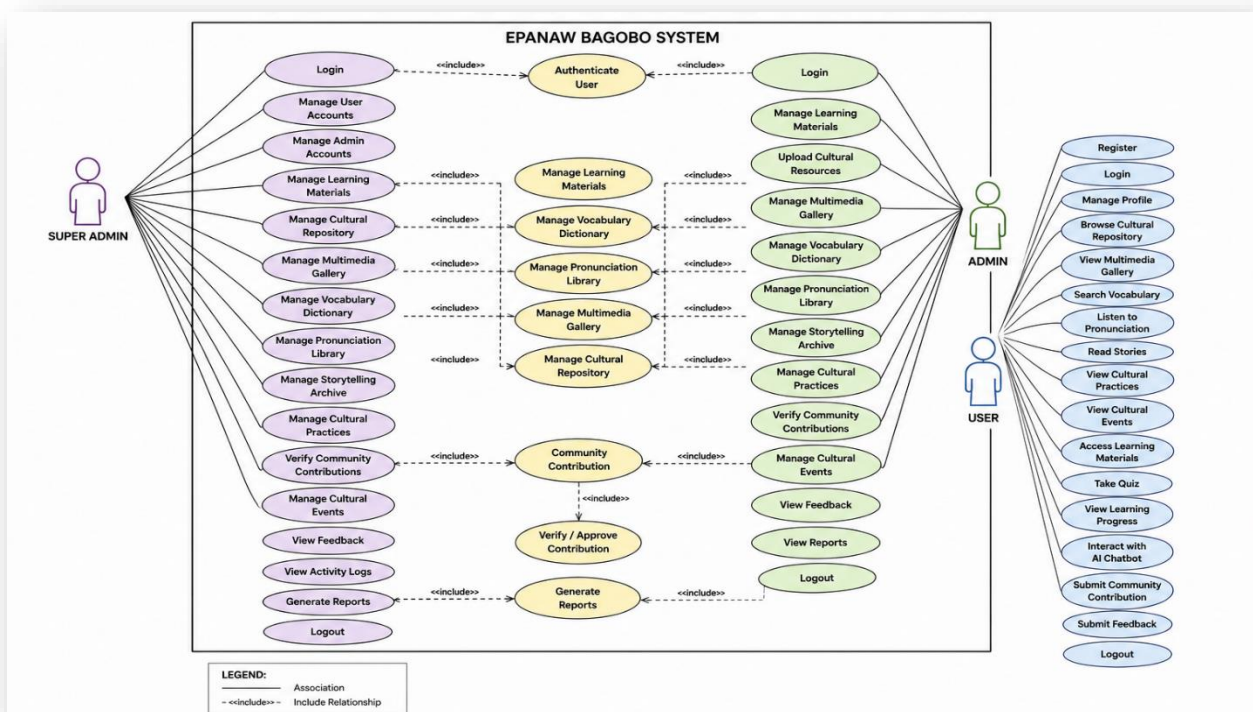


Figure 6: Use Case Diagram of the System

Data Flow Diagram. The Data Flow Diagram (DFD) illustrates the movement of data within the proposed system. It presents how data are received, processed, stored, and transmitted throughout the system. The DFD identifies the interactions between external entities, system processes, and data stores, providing a clear representation of how information flows to support the system's functionalities. It serves as a logical model that ensures efficient data processing, secure information management, and effective communication among the different components of the proposed system. The Data Flow Diagram (DFD) Level 0, provides a high-level representation of the proposed system. It illustrates the interaction between the system and its external entities, namely the User, Admin, and Super Admin, by showing the flow of information entering and leaving the system. As illustrated in the diagram based on the figure below, the User interacts with the system by submitting registration details, login credentials, learning activities, AI chatbot queries, community contributions, and feedback, while receiving learning materials, cultural resources, AI-generated responses, quiz results, and learning progress. The admin manages cultural resources, learning materials, multimedia content, and community contributions, whereas the Super Admin oversees user management, system monitoring, and report generation. Overall, the Context Diagram defines the scope of the proposed system and demonstrates how data are exchanged between the external entities and the system to support its core functionalities.

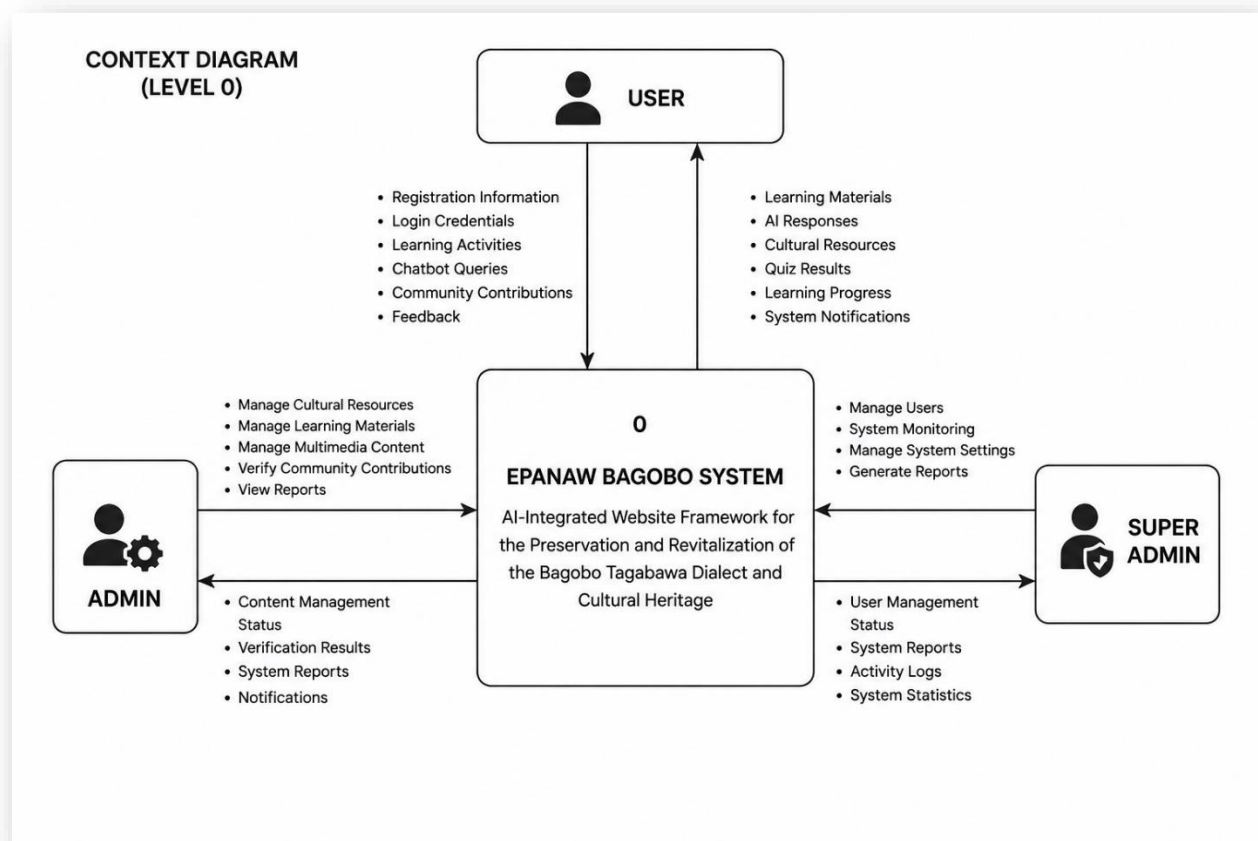


Figure 7: Context Diagram Level 0 of the System

The Data Flow Diagram (DFD) Level 1 that was shown in Figure 8 below presents the major processes of the proposed system. It illustrates how data flow between the external entities, the system's core processes, and the database, ensuring efficient processing, storage, and retrieval of information. As shown in the Figure 8, the system consists of six major processes: User Management, Learning Management, Cultural Repository Management, AI Chatbot Processing, Community Contribution Management, and Feedback Management. Each process communicates with its corresponding database tables to store and retrieve information. User-related data are managed through the User

Accounts table, while learning activities are stored in the Learning Materials, Learning Progress, and Quiz Results tables. Cultural resources are maintained in the Cultural Repository, Multimedia Gallery, Storytelling Archive, Cultural Practices, Vocabulary Dictionary, and Pronunciation Library tables. Community submissions and verification records are stored in the Community Contributions and Resource Verification tables, while chatbot interactions, user feedback, and system activities are recorded in the AI Chatbot Logs, Feedback, and Activity Logs tables. Overall, the DFD Level 1 demonstrates how the system processes user requests and stores the resulting information in the database, ensuring organized data management, efficient retrieval of records, and reliable support for the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

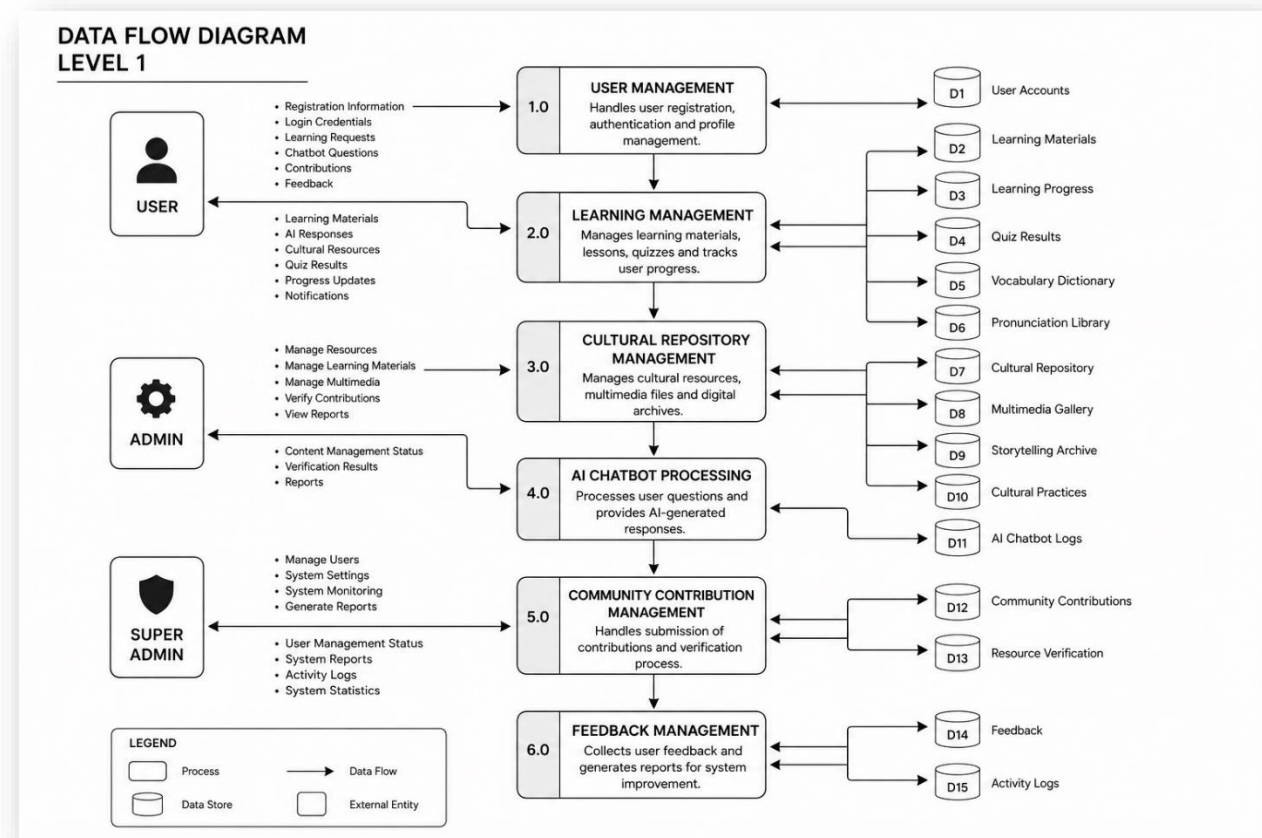


Figure 8: Data Flow Diagram of the System

Operational Framework. The Operational Framework illustrates the overall operation of the proposed system. It presents the relationship between the system's inputs, processes, and outputs, providing a clear overview of how data are collected, processed, and transformed into meaningful information. This framework serves as a guide in understanding the operational flow of the proposed system and how it supports the achievement of the study's objectives. The figure 9 below illustrates the overall operation of the proposed system using the Input–Process–Output (IPO) model. The Input stage consists of data provided by users and administrators, including registration details, login credentials, learning materials, cultural resources, vocabulary entries, pronunciation records, multimedia files, community contributions, AI chatbot queries, and user feedback. The Process stage involves user authentication, Role-Based Access Control (RBAC), learning management, AI chatbot processing, cultural repository management, multimedia management, resource verification, quiz evaluation, database storage, and report generation. These processes ensure that all information is securely managed and efficiently processed within the system. Finally, the Output stage delivers the expected results, including secure user access, an organized digital cultural repository, interactive learning resources, AI-generated responses, quiz results, learning progress, verified community contributions, multimedia content, administrative reports, and the preservation of the Bagobo Tagabawa dialect and cultural heritage. The Operational Framework demonstrates how the proposed system transforms collected data into meaningful information and services that support the objectives of the study.

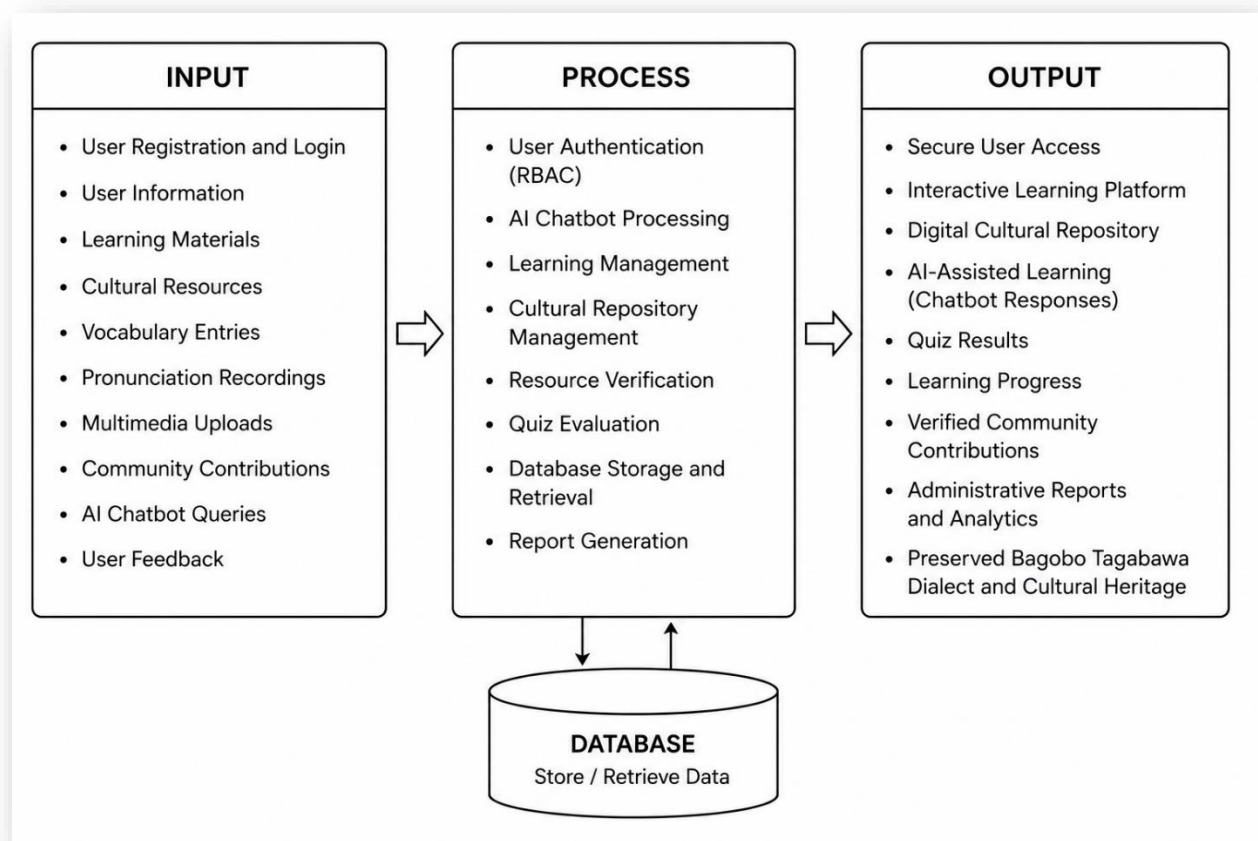


Figure 9: Operational Framework of the Study

Flowchart of the System. The System Flowchart illustrates the overall operational flow of the proposed system. It presents the sequence of processes performed by the system, from user authentication to accessing the system's major features. The flowchart provides a clear visualization of the system's workflow and decision-making processes, demonstrating how users interact with the platform and how information is processed throughout the application. The System Flowchart in Figure !0 below illustrates the overall workflow of the proposed system, beginning when a user accesses the EPANAW BAGOBO website. The user may either register a new account or log in using existing credentials. The system validates the submitted credentials against

the User Accounts database. If the authentication fails, an error message is displayed and the user is prompted to try again. Once authentication is successful, the system identifies the user's role through the implemented Role-Based Access Control (RBAC) mechanism. Based on the assigned role, the Super Admin manages user accounts, monitors system activities, and generates reports; the admin manages learning materials, cultural resources, multimedia content, and verifies community contributions; while the User accesses learning modules, explores the cultural repository, interacts with the AI chatbot, completes quizzes, monitors learning progress, and submits community contributions and feedback. Throughout these operations, the system retrieves and stores information in the corresponding database tables to ensure accurate data management and efficient service delivery. After completing the desired activities, the user logs out, marking the end of the system process.

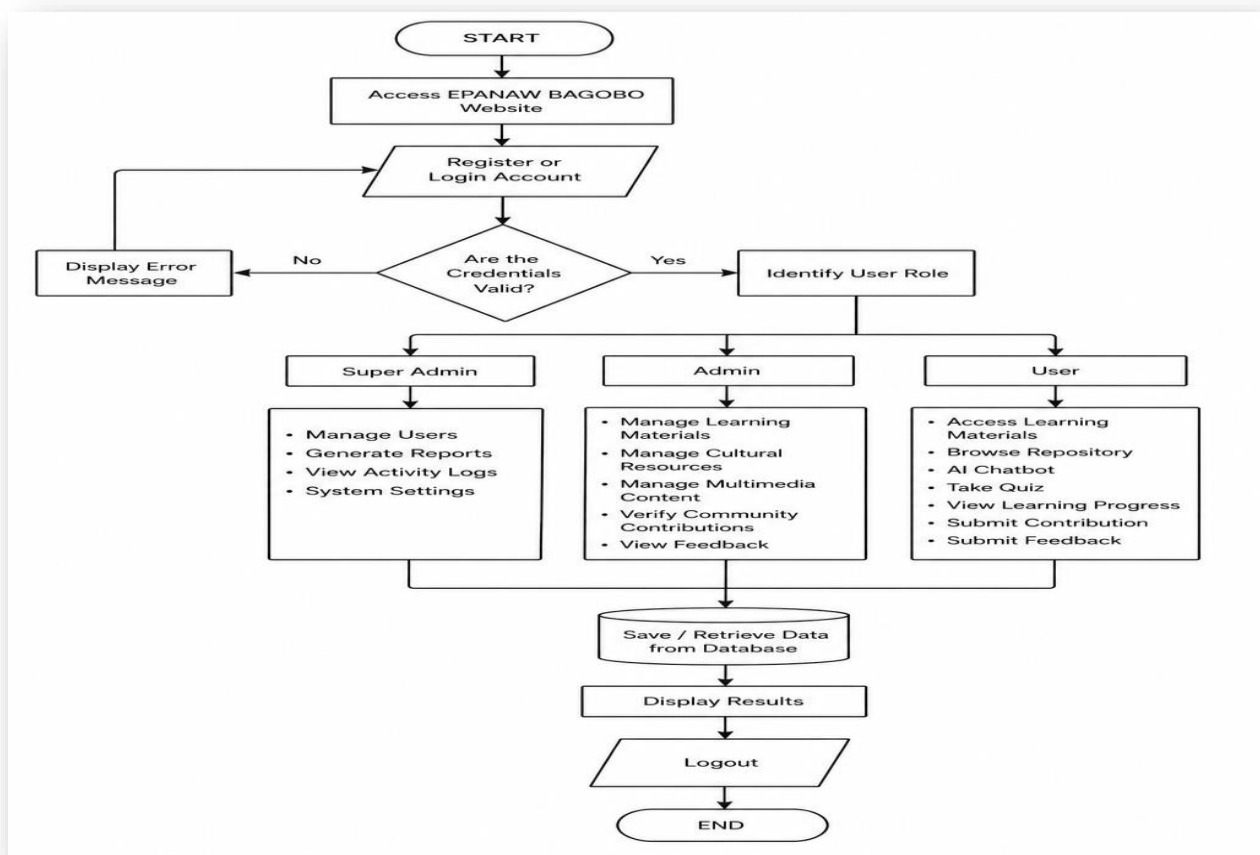


Figure 10: Flowchart of the System

Graphical User Interface Design. The Graphical User Interface (GUI) presents the visual design of the proposed system. It illustrates the layout and appearance of the system's major interfaces, enabling users to interact with the application through a simple, responsive, and user-friendly environment. The GUI is designed to enhance usability by providing intuitive navigation and organized access to the system's learning modules, cultural resources, AI-assisted services, and administrative functions.



Figure 11: Proposed Mock-up user interface of the Web-based Bagobo Tagabawa cultural preservation system

The Landing Page shows in Figure 11 above serves as the entry point of the proposed system. It introduces users to the objectives of the platform and provides quick access to the available features through the navigation menu. It also highlights the importance of preserving the Bagobo Tagabawa dialect and cultural heritage while encouraging users to register or log in to access the system's learning resources and interactive services.

Construction

During the construction phase, the software requirements are transformed into something functional. This process involves coding, testing, and integrating features to meet the project's objectives. The proponent will work closely with stakeholders to clarify requirements, gather feedback, and ensure alignment with goals. The delivery of value will be prioritized while maintaining flexibility to accommodate changes.

Implementation Plan. It outlines the activities required to develop, test, and deploy the system. It serves as a guide for the systematic execution of the project, ensuring that each development phase is completed efficiently and in accordance with the objectives of the study. The implementation process includes system development, database integration, testing, deployment, and maintenance to ensure that the proposed system operates effectively and meets the needs of its intended users. The implementation of the proposed system will follow a structured development process to ensure that all components are properly developed, integrated, and evaluated. The

project begins with the review of the approved system requirements and design specifications, followed by the development of the web application using Laravel, HTML, CSS, JavaScript, and MySQL. After the core modules have been developed, the database will be integrated to support secure data storage and retrieval. The system will then undergo comprehensive testing to verify its functionality, reliability, and usability before deployment. Once validated, the application will be deployed for use by its intended users. Continuous maintenance and updates will be performed to improve system performance, address technical issues, and ensure the long-term effectiveness of the platform in supporting the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

System Development. The proposed system will be developed using Laravel as the primary web application framework, with HTML5, CSS3, Bootstrap, and JavaScript for the user interface, and MySQL as the database management system. It will follow a Three-Tier Architecture consisting of the Presentation, Application, and Data Layers to ensure efficient processing and secure data management. The system will integrate an AI-powered chatbot and include modules for user authentication, learning management, cultural repository management, vocabulary and pronunciation libraries, multimedia gallery, community contributions, feedback, and administrative functions. Each module will be individually tested before integration to ensure system reliability, functionality, and support for the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Development of User Module. It was developed to provide learners and community members with access to the educational and cultural features of the proposed system. Registered users can create and manage their accounts, access learning modules, browse the cultural repository, search the vocabulary dictionary, listen to pronunciation recordings, view multimedia resources, explore the storytelling archive, interact with the AI-powered chatbot, participate in quizzes, monitor their learning progress, submit community contributions, and provide feedback. The module is designed to offer a secure, user-friendly, and interactive learning environment that encourages active participation in preserving and revitalizing the Bagobo Tagabawa dialect and cultural heritage.

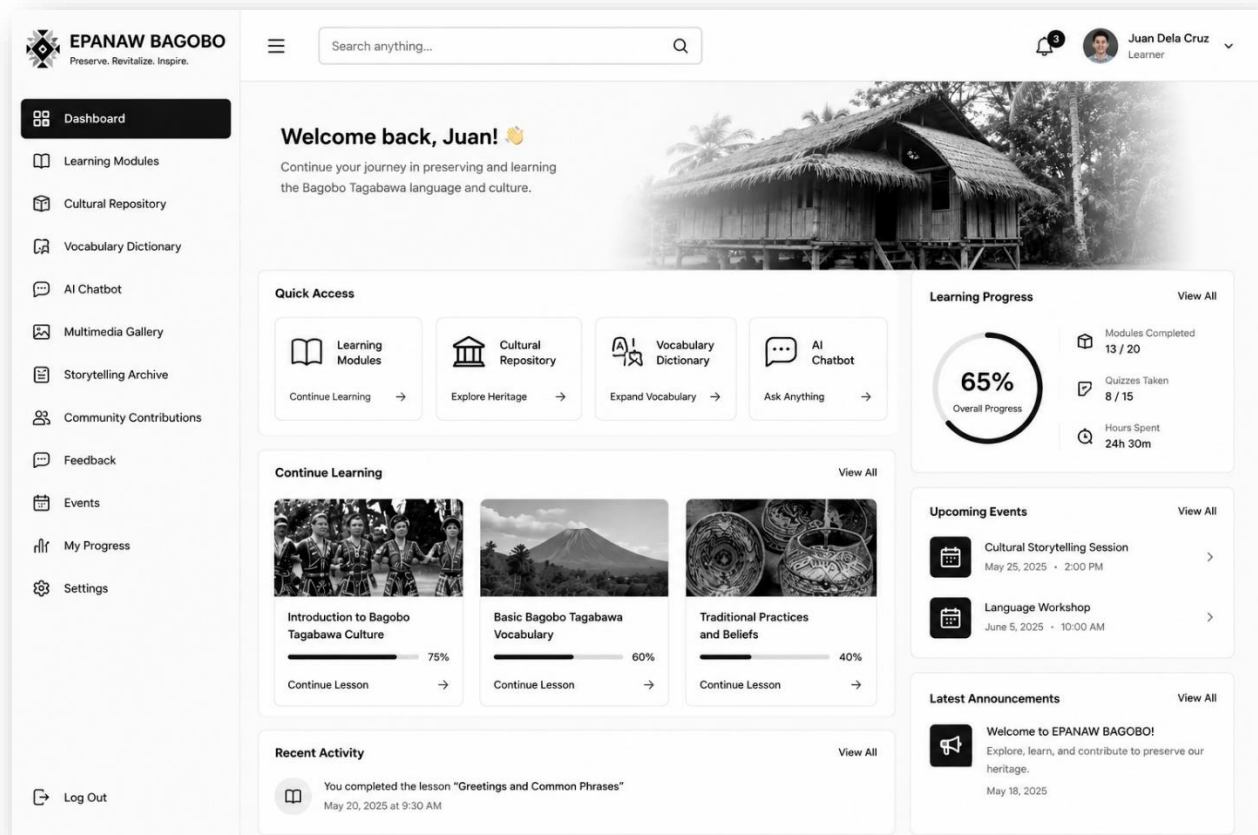


Figure 12: Proposed Mock-up User Module

As shown in the interface in Figure 12 above, the dashboard contains a navigation panel that provides access to the Learning Modules, Cultural Repository, Vocabulary Dictionary, AI Chatbot, Multimedia Gallery, Storytelling Archive, Community Contributions, Feedback, Events, Learning Progress, and Account Settings. The main dashboard displays a personalized welcome message, quick access shortcuts, learning progress statistics, ongoing learning modules, upcoming cultural events, and recent announcements. These features allow users to easily continue their learning activities, explore cultural resources, and stay informed about the latest updates within the platform. The dashboard also integrates with the system database by retrieving user information, learning progress, completed quizzes, cultural resources, and event details from their respective database tables. This enables the system to provide personalized content, monitor user engagement, and support the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage through an organized, interactive, and efficient web-based learning environment.

Development of Admin Module. The Admin Module was developed to manage the educational and cultural content of the proposed system. This module enables administrators to create, update, organize, and delete learning materials, cultural resources, vocabulary entries, pronunciation records, multimedia content, and storytelling archives. Administrators are also responsible for reviewing and verifying community contributions before publication to ensure the accuracy and authenticity of the information available on the platform. In addition, the module provides access to user

feedback, content statistics, and reports that support effective content management and system maintenance. Access to administrative functions is controlled through the system's Role-Based Access Control (RBAC) mechanism.

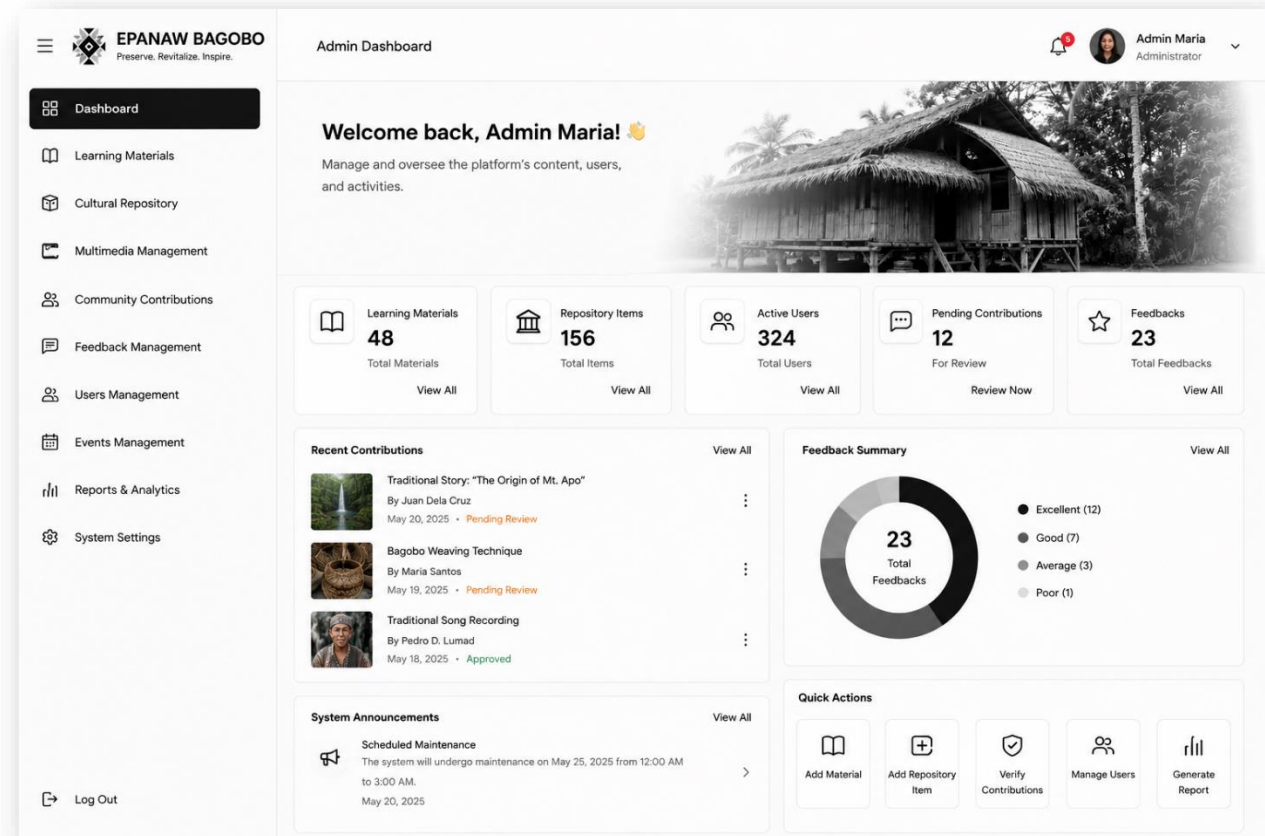


Figure 13: Proposed Mock-up Admin Module

As shown in the interface based on the Figure 13 above, the dashboard provides navigation to the Learning Materials, Cultural Repository, Multimedia Management, Community Contributions, Feedback Management, User Management, Events Management, Reports and Analytics, and System Settings modules. The main dashboard displays summary statistics, including the total number of learning materials, repository items, active users, pending community contributions, and user feedback. It also provides sections for recent community submissions, feedback summaries, system

announcements, and quick-action shortcuts that allow administrators to efficiently perform content management and verification tasks. The Admin Dashboard is integrated with the system database, allowing administrators to retrieve, update, approve, and manage information stored in the corresponding database tables. Through this interface, administrators can maintain learning materials, cultural resources, multimedia content, community contributions, feedback records, and event information. This functionality ensures that the system remains organized, secure, and up to date, thereby supporting the effective preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Development of Super Admin Module. The Super Admin Module was developed to provide complete administrative control over the proposed system. This module enables the Super Administrator to manage user accounts, assign user roles and permissions, monitor system activities, configure system settings, generate reports, and maintain database integrity. It also allows the Super Administrator to oversee the overall operation of the platform, ensuring system security, data consistency, and reliable performance. Through Role-Based Access Control (RBAC), the Super Administrator has full access to all modules and administrative functions within the system.

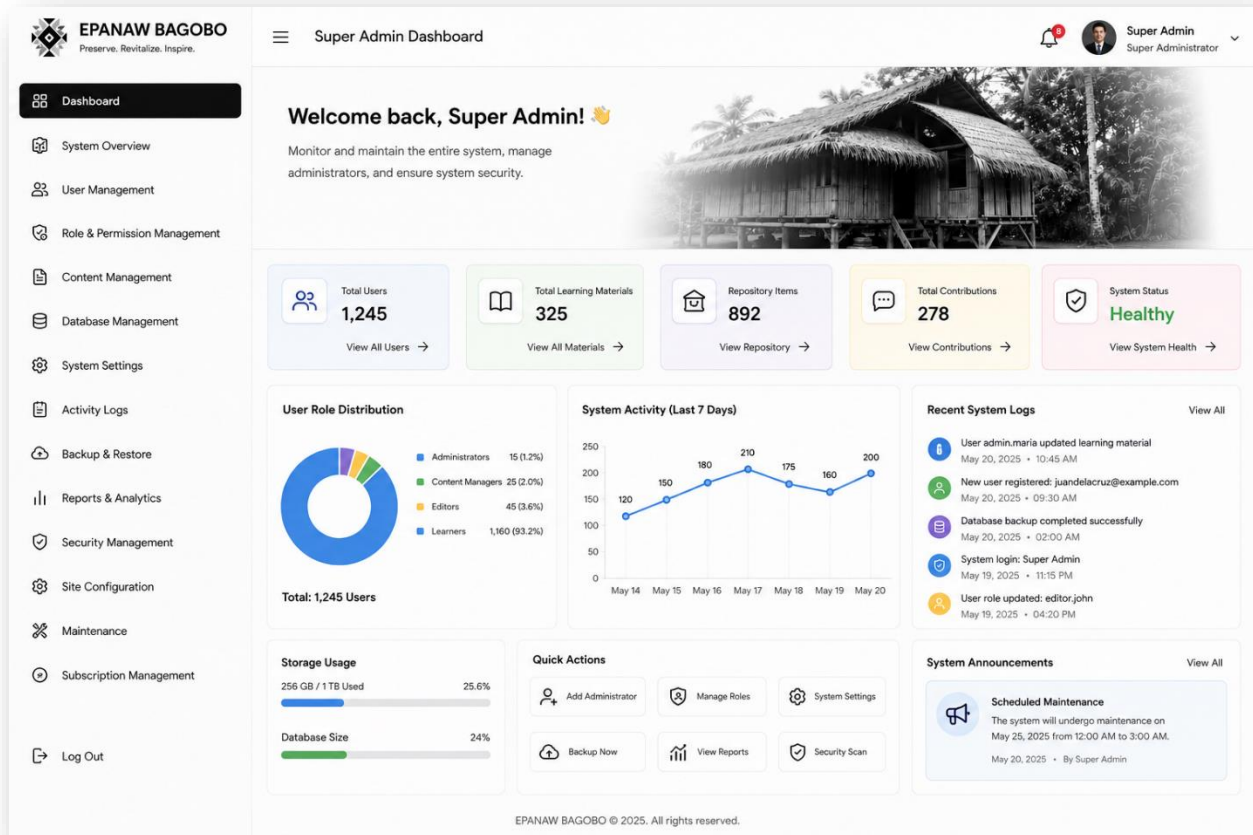


Figure 14: Propose Mock-up Super Admin Module

As shown in the interface, the dashboard includes modules for System Overview, User Management, Role and Permission Management, Content Management, Database Management, System Settings, Activity Logs, Backup and Restore, Reports and Analytics, Security Management, Site Configuration, Maintenance, and Subscription Management. The main dashboard presents key system statistics, including the total number of users, learning materials, repository items, community contributions, and system health. It also displays user role distribution, recent system activities, storage utilization, system announcements, and quick-action tools for managing administrators, generating reports, performing database backups, and configuring system settings. The Super Admin Dashboard is directly integrated with the system database, allowing the Super

Administrator to manage all system records, monitor user activities, maintain database integrity, and oversee administrative operations. This interface ensures efficient system administration, enhances data security, and supports the reliable operation of the proposed platform, contributing to the effective preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

How the Project will Work

The proposed EPANAW BAGOBO system is a web-based platform that allows users to access learning modules, cultural resources, AI-assisted language support, multimedia content, and community features through a secure and user-friendly interface. After registration and login, users are directed to role-specific dashboards where they can perform authorized functions. Administrators manage and verify system content, while the Super Administrator oversees user management, system security, and overall platform operations. The system utilizes a Three-Tier Architecture and a MySQL database to efficiently process, store, and retrieve information, supporting the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Super Admin Page:

1. Log in to the system.
2. Access the Super Admin Dashboard.
3. Manage users, roles, and system settings.
4. Monitor system activities and generate reports.
5. Log out securely.

Admin Page:

1. Log in to the system.
2. Access the Admin Dashboard.
3. Manage learning materials, cultural resources, and multimedia content.
4. Review and verify community contributions and feedback.
5. Save updates and log out.

User Page:

1. Register or log in to the system.
2. Access the User Dashboard.
3. Use learning modules, cultural resources, AI chatbot, vocabulary, and multimedia features.
4. Complete quizzes, submit community contributions, and provide feedback.
5. View learning progress and log out.

Deployment Plan. The Deployment Plan describes the implementation of the proposed system of the study after the successful completion of system development and testing. The system will be deployed using Laravel as the web application framework and MySQL as the database management system, with the necessary server components configured to support its operation. After deployment, the database and user accounts will be configured, and all system modules will be verified to ensure proper functionality.

Once validated, the system will be made available to its intended users. Regular maintenance, database backups, and system updates will be conducted to ensure reliable performance, data security, and continuous support for the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Installation Plan. It outlines the procedures for setting up the proposed system. The required software components, including Laravel, PHP, Composer, Apache, and MySQL, will be installed and configured to establish the development and deployment environment. After installing the required software, the database will be created based on the designed schema, and the application files will be configured to establish a secure connection with the database.

Step 1: Install the required software components, including PHP, Composer, Apache Web Server, and MySQL, to establish the development and server environment needed to run the Laravel application.

Step 2: Install the Laravel framework using Composer and configure the application settings, environment variables, and project dependencies to prepare the system for development and deployment.

Step 3: Create the MySQL database and configure its settings based on the designed database schema. This step establishes the database that will store user accounts, learning materials, cultural resources, and other system data.

Step 4: Import the database tables, relationships, and initial records into MySQL to ensure that the system has the necessary structure and sample data for proper operation.

Step 5: Update the Laravel environment configuration file (.env) by specifying the database name, username, password, and host to establish a secure connection between the application and the MySQL database.

Step 6: Perform installation verification by testing the major system modules, including user authentication, learning management, AI chatbot, cultural repository, multimedia gallery, community contributions, and administrative functions to ensure that they operate correctly.

Step 7: Once the installation and verification processes are successfully completed, the system is deployed and made available to the intended users. Regular maintenance, database backups, and system updates are then performed to ensure continuous performance, security, and reliability.

Maintenance Plan. This outlines the activities required to ensure the continuous operation, security, and performance of the proposed system of this study after its deployment. It includes regular system monitoring, software updates, database maintenance, and technical support to ensure that the system remains reliable and responsive to the needs of its users.

Step 1: Monitor the performance, availability, and functionality of the system to detect and resolve issues promptly.

Step 2: Perform regular backups of the database to prevent data loss and ensure successful recovery when necessary.

Step 3: Install system updates, security patches, and framework upgrades to improve performance and maintain compatibility.

Step 4: Update learning materials, cultural resources, multimedia content, and vocabulary entries to keep the information accurate and relevant.

Step 5: Manage user accounts, roles, permissions, and access rights to maintain system security.

Step 6: Resolve reported errors, troubleshoot system issues, and provide assistance to users when needed.

Step 7: Implement improvements and additional features based on user feedback and system evaluation results.

Backup and Recovery Plan. The system outlines the procedures for protecting and restoring the data of the proposed system of the study. It ensures that critical system information can be recovered in the event of hardware failure, software errors, accidental deletion, or other unforeseen incidents. This plan helps maintain data integrity, system availability, and the continuous operation of the proposed system.

Cutover

The proposed system will adopt a Direct Cutover strategy, where the existing manual process will be replaced by the EPANAW BAGOBO web-based system after successful testing and validation. Before the transition, the system will undergo final verification to ensure that all modules, including user authentication, learning management, cultural repository, AI chatbot, multimedia gallery, vocabulary and pronunciation libraries, community contributions, and administrative functions, are operating correctly. The database will be initialized with the required data, and user accounts for the Super Administrator, Administrator, and registered users will be configured. Once the system is verified, it will be deployed and made available for use. During the initial implementation period, the development team will monitor system performance, resolve technical issues, and provide user support to ensure a smooth transition. This approach enables users to immediately benefit from the system's features while supporting the efficient preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage.

Testing and Evaluation

The Testing and Evaluation phase aims to verify the functionality, reliability, and overall quality of the proposed system of the study. This phase ensures that the developed system meets its functional and non-functional requirements before

deployment. It also evaluates the system to determine its effectiveness, usability, performance, security, and overall acceptability among its intended users.

Testing. This phase ensures that the functions according to its specified requirements. During this phase, each system module will be tested individually and as an integrated application to verify its functionality, accuracy, and reliability. The testing process includes validating user authentication, role-based access control, learning modules, cultural repository, AI-powered chatbot, vocabulary dictionary, pronunciation library, multimedia gallery, community contributions, feedback management, and administrative functions. Any identified errors or inconsistencies will be corrected to ensure that the system performs efficiently and provides a stable, secure, and user-friendly experience before deployment.

Evaluation. This phase assesses the overall quality and acceptability of the proposed system after successful testing. The system will be evaluated by selected respondents, including IT experts, faculty members, and end users, using the ISO/IEC 25010 Software Quality Model. The evaluation will measure the system based on Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The collected responses will be analyzed using the weighted mean to determine the system's level of acceptability and identify areas for improvement. The evaluation results will serve as the basis for validating that the proposed system effectively supports the preservation and revitalization of the Bagobo Tagabawa dialect and cultural heritage while meeting the quality standards expected of a web-based application.

Respondents of the Study. The respondents of the study will consist of IT experts, faculty members, and end users selected through purposive sampling. IT experts and faculty members will evaluate the technical quality and functionality of the proposed system, while end users, including members of the Bagobo Tagabawa community, students, and educators, will assess its usability, accessibility, and overall user experience. The evaluation will be conducted and the results will be used to determine the system's overall acceptability and identify areas for improvement. The respondents of the study will consist of 20 participants selected through purposive sampling. The respondents will include three (3) IT experts, five (5) faculty members, and twelve (12) end users, composed of members of the Bagobo Tagabawa community, students, educators, elders, and individuals interested in learning the Bagobo Tagabawa dialect and cultural heritage. IT experts and faculty members will evaluate the technical quality and software performance of the proposed system, while the end users will assess its usability, accessibility, and overall user experience. The results will be used to determine the overall acceptability of the proposed system.

Table 36: Respondent's Distribution

Respondent Group	Number of Respondents
IT Experts	3
Faculty Members	5
End Users (Bagobo Tagabawa community members, students, educators, and interested users)	12
Total	20

Sampling and Design Technique. This study will use a purposive sampling technique to select the respondents who will evaluate the proposed system of the study.

Purposive sampling is appropriate because the respondents will be chosen based on their knowledge, experience, and relevance to the objectives of the study.

Data Gathering Procedure. The data gathering procedure of this study will be conducted in a systematic manner to ensure the accuracy and reliability of the information collected. First, the researchers will secure the necessary permissions from the concerned institutions and the National Commission on Indigenous Peoples (NCIP) before conducting the study. After obtaining approval, the researchers will gather data through interviews, observations, and document reviews involving Bagobo Tagabawa elders, community members, and other reliable sources. The collected information will include Bagobo Tagabawa vocabulary, cultural practices, traditions, stories, audio recordings, images, and other cultural resources that will serve as the content of the proposed system. Once the system has been developed, it will be evaluated by three (3) IT experts, five (5) faculty members, and twelve (12) end users using a questionnaire. The collected responses will be analyzed to determine the functionality, usability, reliability, and overall acceptability of the proposed system. The results of the evaluation will serve as the basis for identifying necessary improvements before the final implementation of the system.

Survey Instrument of the Study. The survey instrument used in this study is a researcher-made questionnaire based on the ISO/IEC 25010 Software Quality Model.

The questionnaire is designed to evaluate the quality and acceptability of the proposed system of the study.

Likert Scale for Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

The study will use a five-point Likert scale to measure the respondents' perceptions of the proposed system. The instrument is based on the ISO/IEC 25010 Software Quality Model, which evaluates the system in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Respondents will indicate their level of agreement with each statement using a scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The responses will be analyzed using the weighted mean to determine the overall quality and acceptability of the proposed system.

Table 37: Likert Scale of the System

Mean Range	Descriptive Equivalent	Interpretation
4.51 – 5.00	Strongly Agree (SA)	The system demonstrates excellent quality and fully satisfies the specified software quality characteristic.
3.51 – 4.50	Agree (A)	The system demonstrates good quality and satisfies most of the specified software quality characteristic, with only minor improvements needed.
2.51 – 3.50	Fairly Agree (FA)	The system demonstrates satisfactory quality but requires further improvements to better satisfy the specified software quality characteristic.
1.51 – 2.50	Disagree (D)	The system demonstrates limited quality and requires significant improvements to satisfy the specified software quality characteristic.
1.00 – 1.50	Strongly Disagree (SD)	The system does not satisfy the specified software quality characteristic and requires major improvements before implementation.

Statistical Tools. The study will use appropriate statistical tools to analyze and interpret the data collected from the respondents. Frequency and Percentage Distribution will be used to describe the respondents' profile, while the Weighted Mean will be used to evaluate the proposed system based on the ISO/IEC 25010 Software Quality Model. These statistical tools will help determine the overall quality, acceptability, and performance of the system of the study.

Equation to calculate the following equations:

1. Percentage and Frequency

$$\text{Percentage} = f/n \times 100$$

Where:

f = Frequency of responses

N = Total number of respondents

2. Weighted Mean

$$\bar{X} = \frac{\sum fx}{N}$$

Where:

$\bar{X}$ = Weighted Mean

f = Frequency of each response

x = Weight assigned to each response (1–5)

N = Total number of respondents

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